

Lecture 04: Generalized linear models

Prof. Jochen Braun Ph.D.

Institute of Biology

Otto-von-Guericke University Magdeburg

Textbook:

B Efron & T Hastie (2016) Computer Age Statistical Inference,

Cambridge University Press.

Abstract:

Regression analysis, either in its classical in modern formulations, exploits the covariance of different variables x and y and attempts to establish some sort of geometrical relation ship. Regression is the statistician's most powerful tool for relating "other" variables to a variable of primary interest

We **revisit linear regression** and present its more general formulation for multiple independent variables x and one dependent variable y . **Ridge regression** can be useful when the number of x -variables get's large.

Generalized linear models take advantage of **known distributions** from the exponential family. They provide exact maximum likelihood results even when our observations are **not normally distributed**.

Logistic regression is the most intuitive generalization and covers situations with **binomially distributed observations** (e.g., counts of successes and failures)

The same principle works for other distributions from the exponential family, such as **Gamma distributions** or **Poisson distributions**.

Overview:

1. Linear regression
2. Ridge regression

3. Logistic regression

4. Generalized linear models

5. Gamma regression

6. Poisson regression

✓ 1. Linear regression

We make scalar observations y_n over N trials, and collect these observations in N -dimensional vector $\mathbf{y}$ ($N \times 1$). We assume that these observations are linearly modulated by experimental factors under our control. Specifically, we devise a **design matrix**, denoted $\mathbf{X}$ ($N \times P$), that structures our experiment such that the experimental factors occur equally (or comparably) often in different combinations.

Our linear model is

$$\mathbf{y} = \mathbf{X} \cdot \boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2)$$

where $\boldsymbol{\beta} = \{\beta_1, \dots, \beta_P\}$ is a $(P \times 1)$ vector of unknown parameters and $\boldsymbol{\epsilon} = \{\epsilon_1, \dots, \epsilon_N\}$ is independent random noise with mean 0 and variance σ^2 .

The design matrix $\mathbf{X}$ must be chosen in such a way that the inner product matrix $\mathbf{S}$ is invertible!

$$\mathbf{S} = \mathbf{X}^T \cdot \mathbf{X}, \quad \mathbf{S}^{-1} \cdot \mathbf{S} = \mathbf{I}$$

The least-squares estimate, which goes back to Gauss and Legendre in the early 18th century, is the parameter vector that minimizes the total sum of squared errors

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta}\|^2$$

In other words, we wish to minimize the objective function

$$\mathcal{L}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta})^T \cdot (\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta})$$

with respect to $\boldsymbol{\beta}$. Note that we can expand the sum of squares in terms of a dot product $(1 \times N) \cdot (N \times 1)$

Setting the derivative with respect to $\boldsymbol{\beta}$ to zero

$$0 \stackrel{!}{=} \frac{\partial \mathcal{L}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \frac{\partial}{\partial \boldsymbol{\beta}} [(\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta})^T \cdot (\mathbf{y} - \mathbf{X} \cdot \boldsymbol{\beta})] =$$

$$\begin{aligned}
 \frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}} &= \frac{\partial}{\partial \boldsymbol{\beta}} [\mathbf{y}^T \cdot \mathbf{y} - 2\boldsymbol{\beta}^T \cdot \mathbf{X}^T \cdot \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \cdot \mathbf{X} \cdot \boldsymbol{\beta}] = \\
 &= -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \cdot \mathbf{X} \cdot \boldsymbol{\beta}
 \end{aligned}$$

and solving for $\boldsymbol{\beta}$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \cdot \mathbf{X})^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y} = \mathbf{S}^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}, \quad \mathbf{S} \equiv \mathbf{X}^T \cdot \mathbf{X}$$

These are called the **normal equations** and $\mathbf{S}$ is the $(P \times P)$ **inner product matrix**.

$\hat{\boldsymbol{\beta}}$ is an unbiased estimate for $\boldsymbol{\beta}$ and has the covariance matrix $\mathbf{S}^{-1} \sigma^2$

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta}, \mathbf{S}^{-1} \sigma^2)$$

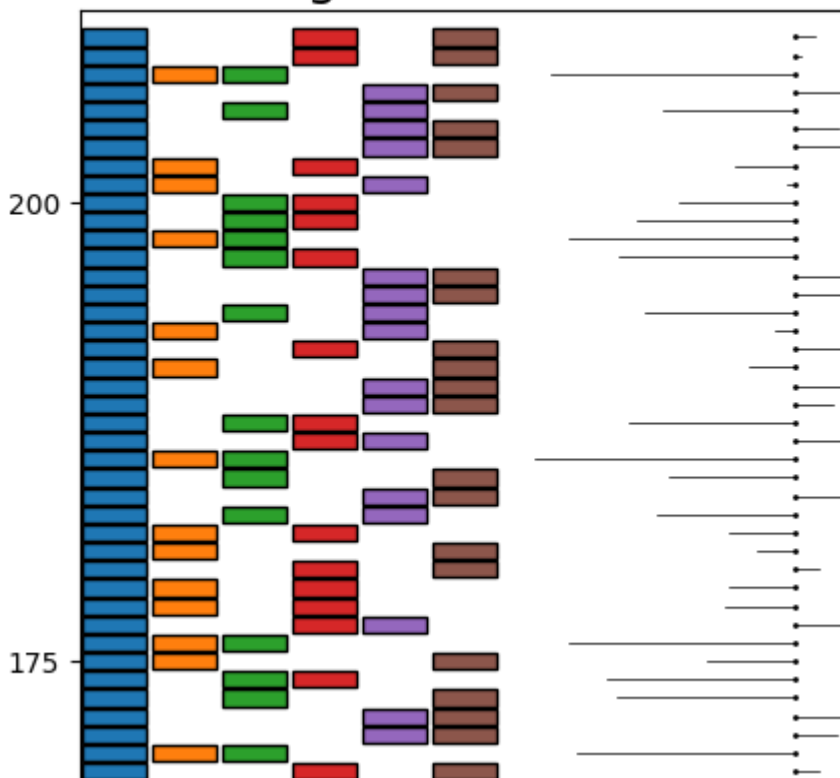
Thus, the i -th diagonal element of the covariance matrix estimates the variance of the individual β_i .

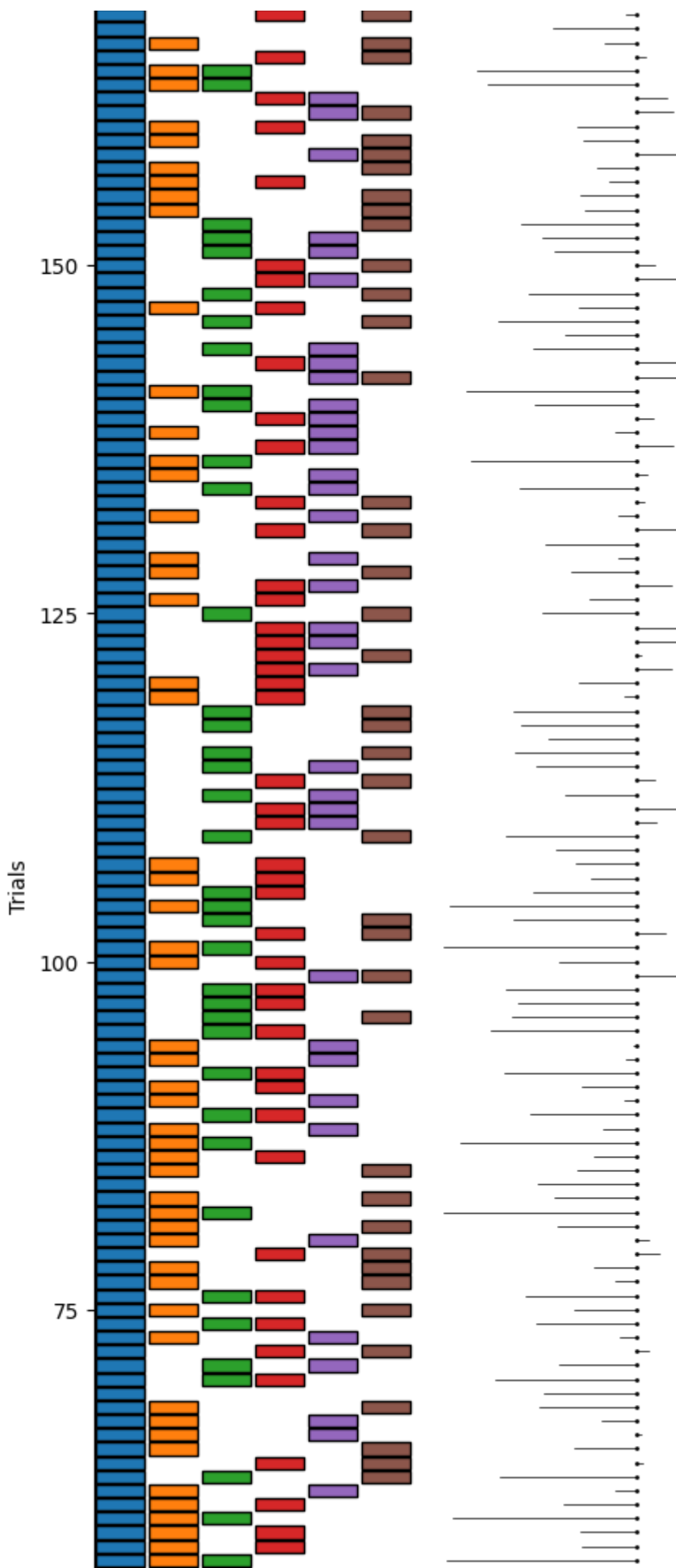
Linear regression

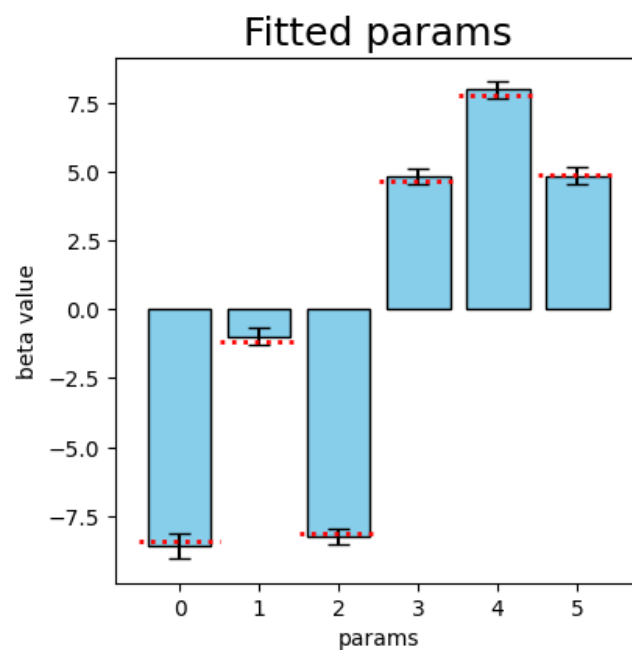
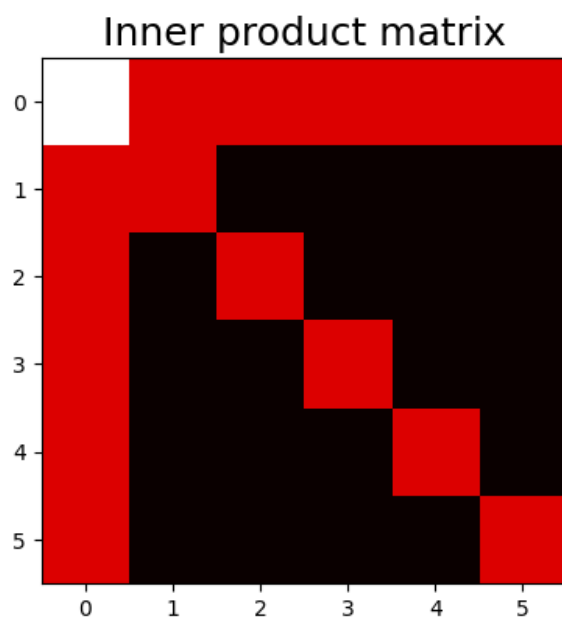
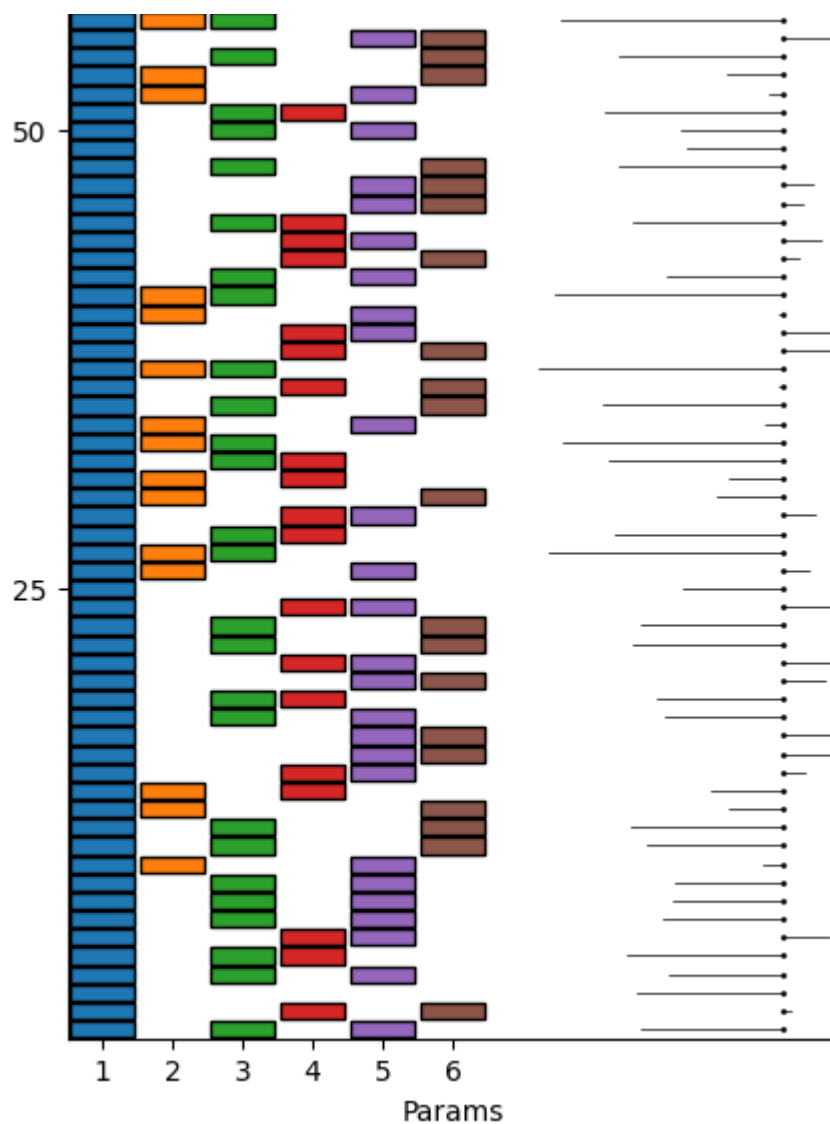
Show code

Pparam 6
 Kcombo 2
 Nrepeat 10
 Sigma 1.50

Design matrix & data







plot_linear_regression

```
def plot_linear_regression(Pparam=6, Kcombo=2, Nrepeat=10, Sigma=0.5)
```

(Slide-25)

Linear regression, alternative code from Grok

Show code

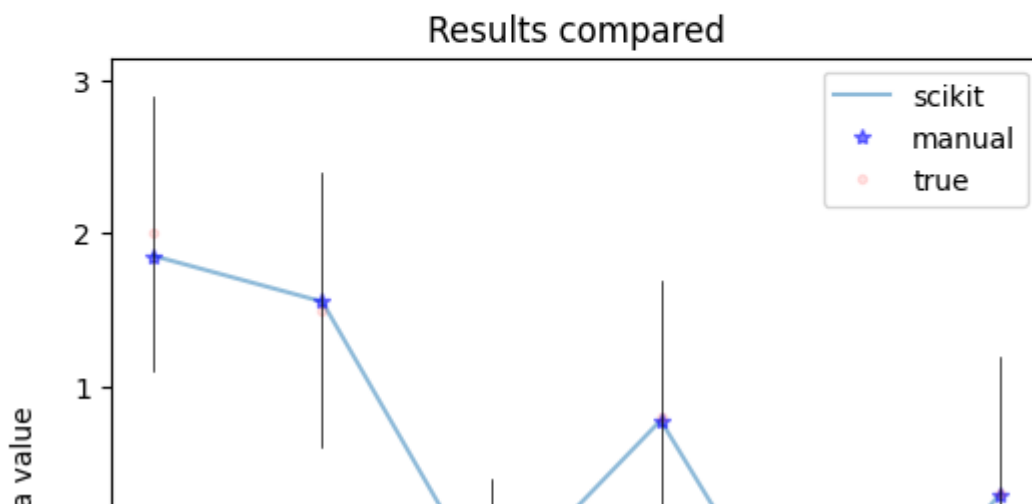
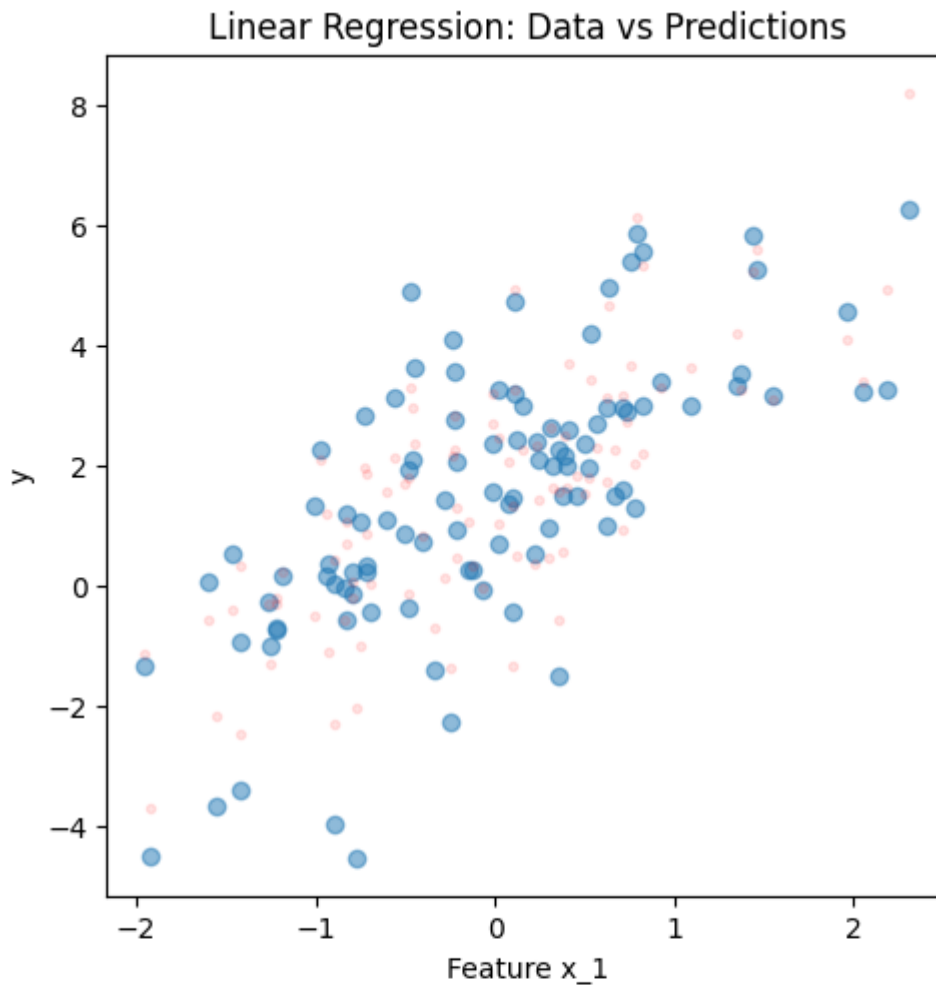
```
(6, 6)
```

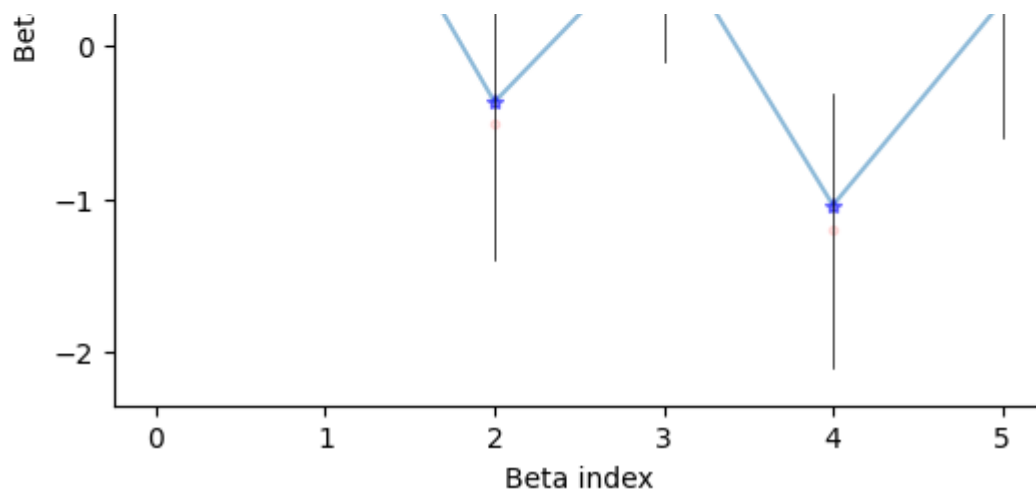
```
True parameters: [ 2.   1.5 -0.5  0.8 -1.2  0.3]
```

```
Estimated parameters (scikit-learn): [ 1.85387526  1.55421281 -0.36448897
```

```
Estimated parameters (manual): [ 1.85387526  1.55421281 -0.36448897  0.77
```

```
Mean Squared Error: 0.8033
```





2. Ridge regression

When the number of predictors is large, we are often looking for a few interesting predictor variables in a haystack of uninteresting ones. The prior belief is that most of the β values lie near zero. We can take this into account with **ridge regression**, which biases the maximum likelihood estimates $\hat{\beta}$ toward zero.

The downside is that ridge regression attenuates predictors that are genuinely outstanding (in a positive or negative sense). So if we do find some particularly strong predictors, their impact will likely be **underestimated**.

Specifically, we minimize the modified cost function

$$\begin{aligned}\mathcal{L}(\beta, \lambda) &= \|\mathbf{y} - \mathbf{X} \cdot \beta\|^2 + \lambda \|\beta\|^2 = \\ &= (\mathbf{y} - \mathbf{X} \cdot \beta)^T \cdot (\mathbf{y} - \mathbf{X} \cdot \beta) + \lambda \beta^T \cdot \beta\end{aligned}$$

where λ is the **ridge parameter**.

Setting the derivative with respect to β to zero and solving for β now produces

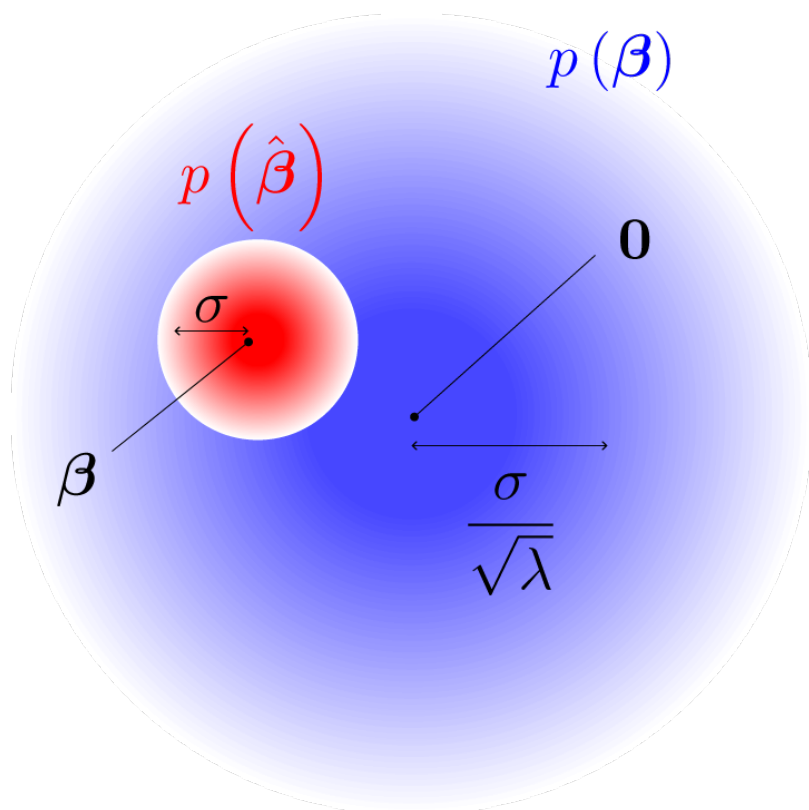
$$0 = -2 \mathbf{X}^T \mathbf{y} + 2 \mathbf{X}^T \cdot \mathbf{X} \cdot \beta + 2 \lambda \beta$$

$$\hat{\beta}(\lambda) = (\mathbf{S} + \lambda \mathbf{I})^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}$$

where $\mathbf{S} = \mathbf{X}^T \cdot \mathbf{X}$ is the inner product matrix.

Comparing $\hat{\beta}(\lambda)$ to the underlying $\beta = \mathbf{S}^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}$, we see that

$$\hat{\beta}(\lambda) = (\mathbf{S} + \lambda \mathbf{I})^{-1} \cdot \mathbf{S} \cdot \beta$$



This corresponds to the conditional expectation of $\hat{\beta}(\lambda)$, given β , assuming normally distributed noise and

$$\hat{\beta}(\lambda) \sim \mathcal{N}(\beta, \sigma^2 \mathbf{I}), \quad \beta \sim \mathcal{N}\left(0, \frac{\sigma^2}{\lambda} \mathbf{I}\right)$$

Thus, ridge regression amounts to a prior belief that the β values lie near zero.

Practicalities

As our predictor variables may come on different scales, we normalize each column of the design matrix $\mathbf{X}$ to zero mean and unit sum of squares. This ensures that all β values are on comparable scales. In addition, we subtract the mean from our observations $\mathbf{y}$

$$\begin{aligned} \sum_i X_{ij} &= 0, & \sum_i X_{ij}^2 &= 1, & \forall j \\ \sum_i y_i &= 0 \\ S_{jk} &= \sum_i X_{ij} X_{ik}, & S_{jj} &= 1 \end{aligned}$$

With these normalizations, the covariance matrix of $\hat{\beta}$ follows from

$$\hat{\beta} = (\mathbf{S} + \lambda \mathbf{I})^{-1} \cdot \mathbf{X}^T \mathbf{y}$$

as

$$\text{Cov}_\lambda = (\mathbf{S} + \lambda \mathbf{I})^{-1} \cdot \mathbf{S} \cdot (\mathbf{S} + \lambda \mathbf{I})^{-1}$$

The diagonal elements of this matrix provide the variance of our estimate $\hat{\beta}(\lambda)$

✓ Diabetes example

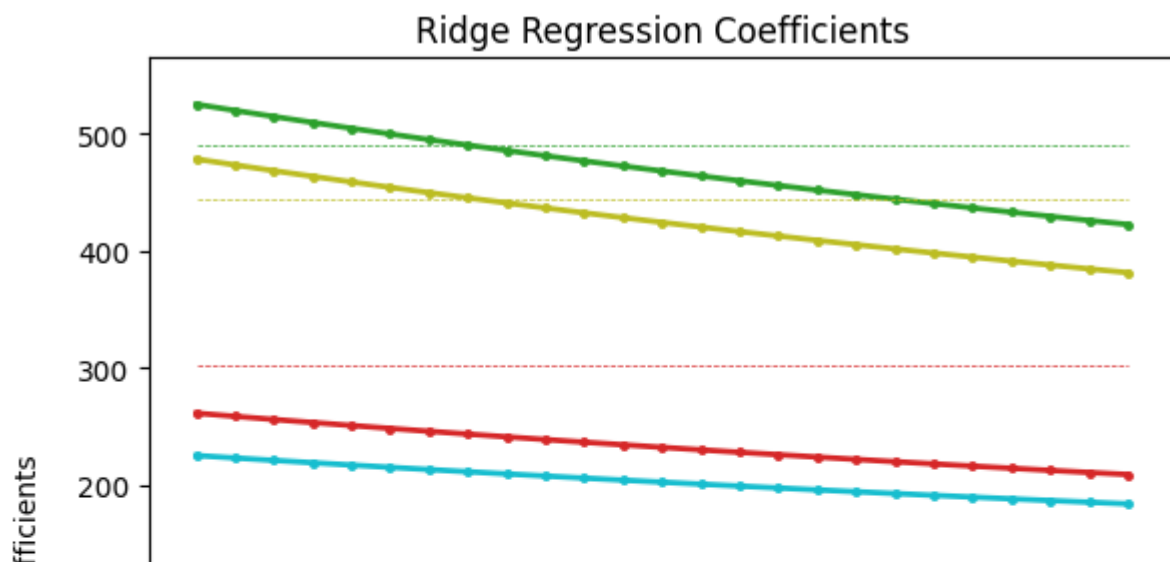
In a diabetes study, 442 patients were assessed with respect to *age*, *sex*, *bmi* (body mass index), *map* (mean arterial blood pressure), and six blood serum measurements, *tc*, *ldl*, *hdl*, *tch*, *ltg*, *glu*. Additionally, disease progression *prog* was established one year after these measurements. We wish to know the comparative predictiveness of these ten measures for disease progression.

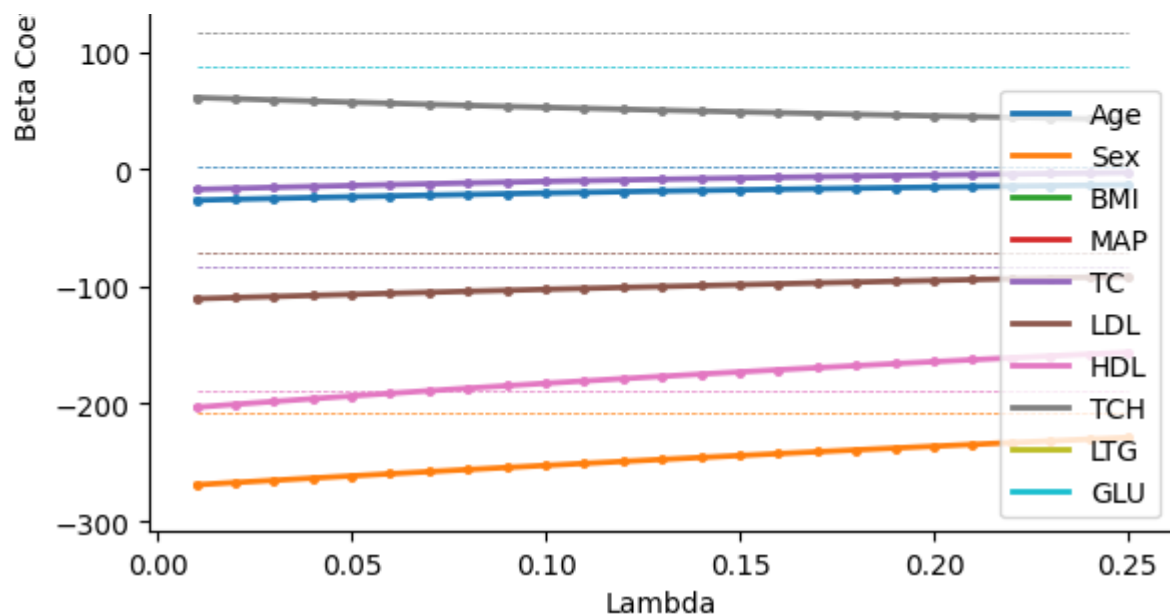
Table 7.2 *First 7 of $n = 442$ patients in the diabetes study; we wish to predict disease progression at one year “prog” from the 10 baseline measurements **age**, **sex**, ..., **glu**.*

age	sex	bmi	map	tc	ldl	hdl	tch	ltg	glu	prog
59	1	32.1	101	157	93.2	38	4	2.11	87	151
48	0	21.6	87	183	103.2	70	3	1.69	69	75
72	1	30.5	93	156	93.6	41	4	2.03	85	141
24	0	25.3	84	198	131.4	40	5	2.12	89	206
50	0	23.0	101	192	125.4	52	4	1.86	80	135
23	0	22.6	89	139	64.8	61	2	1.82	68	97
36	1	22.0	90	160	99.6	50	3	1.72	82	138
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

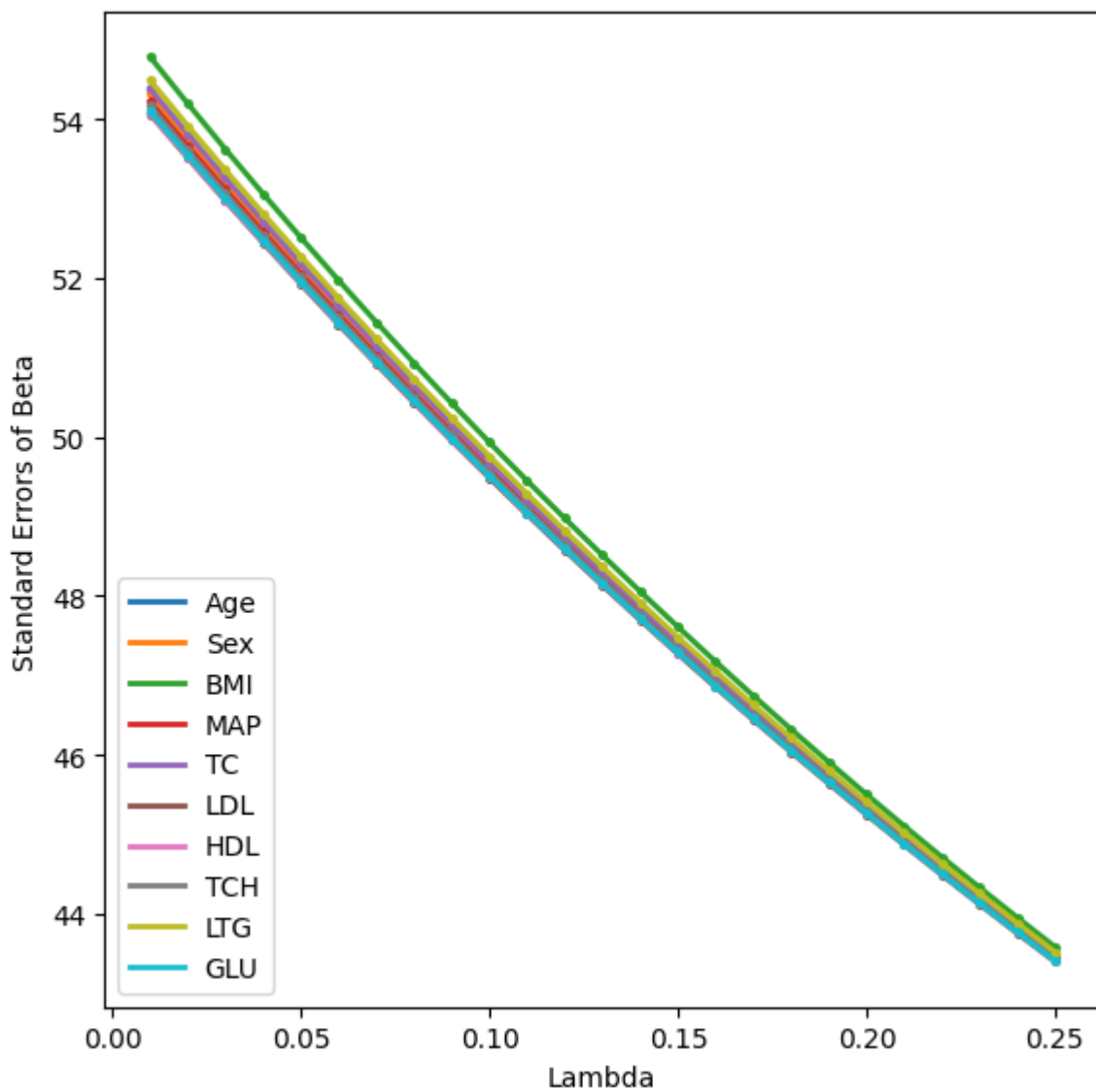
Ridge regression, diabetes example

Show code





Standard Errors vs. Lambda



```

Beta (true): [  1.3 -207.2  489.7  301.8  -83.5  -70.8 -188.7  115.7  44
Beta (SVD): [ -27.45750928 -271.09149264  529.97743612  264.15444782 -18
             -111.53530974 -205.64703198   61.84849271  483.07805804  227.43801714]

```

Conclusion

When the number of predictors is large, it can be helpful to introduce a prior for predictors to be near zero. This reduces the variability of predictor estimates, but does not guarantee that the ridged prediction

$$\mu(\lambda) = \mathbf{X} \cdot \hat{\beta}(\lambda)$$

is better than the ordinary least-squares prediction

$$\mu = \mathbf{X} \cdot \hat{\beta}$$

Note that there is no theoretical guidance for choosing the ridge parameter λ , so that this choice is a matter of experience. For a normalized design matrix $\mathbf{X}$ (column sum is zero, column sum of squares is unity), values between 0 and 0.25 are conventional.

3. Logistic regression

Imagine that we conduct a bioassay or dose-response experiment, where we inject mice with something nasty in different doses. In this case, we would observe the proportion of surviving mice as a function of dose.

Or we could perform a behavioural experiment that counts the proportion of successes and failures as a function of stimulus condition (task difficulty).

In such cases, our data are discrete counts or proportions: survivors of some number of animals, successes in some number of trials, and so on. Logistic regression is a specialized technique for handling such observations.

We define a **logit parameter** $\lambda \in [-\infty, +\infty]$ as a logarithmic function of probability $\pi \in [0, 1]$

$$\lambda = \ln\left(\frac{\pi}{1 - \pi}\right)$$

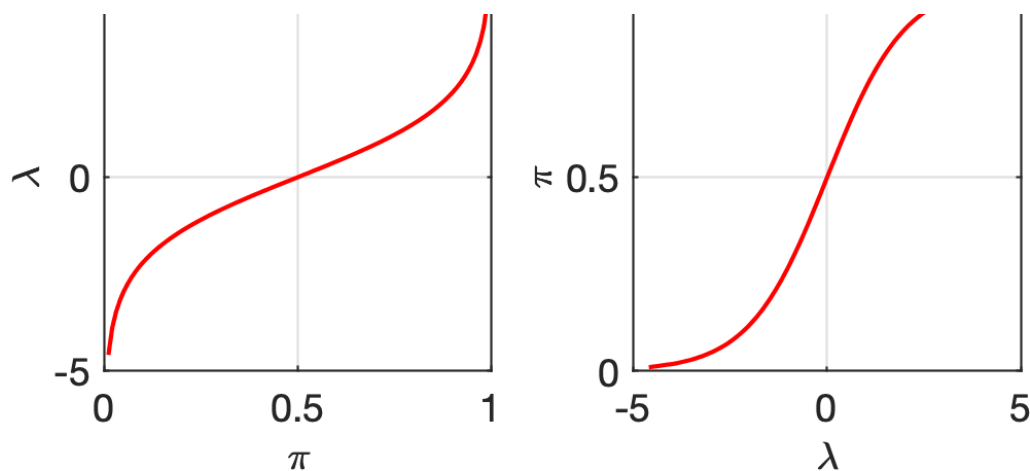
and assume that it depends linearly on predictors x

$$\lambda = a_0 + a_1 x_1 + a_2 x_2 + \dots$$

Consequences of this definition are

$$e^\lambda = \frac{\pi}{1 - \pi} \quad \Leftrightarrow \quad \pi = \frac{e^\lambda}{1 + e^\lambda} = \frac{1}{1 + e^{-\lambda}}, \quad 1 - \pi = \frac{1}{1 + e^\lambda}$$





The term **logistic regression** reflects this use of a **logistic function** for mapping between real numbers $\lambda \in [-\infty, +\infty]$ and probabilities $\pi \in [0, 1]$.

Maximum likelihood fitting

Suppose we have observed y_i 'successes' in n_i trials, where index i runs over N experimental 'conditions'. Suppose further that each 'condition' is associated with several 'predictors' x_{ij} , where index j runs over the number of predictors.

Our model assumes a linear dependence of the logit parameter on predictors x_{ij}

$$\lambda_i = a_0 + a_1 x_{i1} + \dots, \quad \Leftrightarrow \quad \pi_i = \left[1 + e^{-(a_0 + a_1 x_{i1} + \dots)} \right]^{-1}$$

with the derivatives

$$\frac{\partial \lambda_i}{\partial a_0} = 1, \quad \frac{\partial \lambda_i}{\partial a_1} = x_{i1}, \quad \frac{\partial \lambda_i}{\partial a_2} = x_{i2}, \quad \dots$$

Next we connect the π 's of this model to the likelihood of different possible observation. The structure of our data -- integer numbers of successes in an integer number of trials -- imposes a **binomial distribution**. For example, if there are n_i trials in 'condition' i and the proportion of successes is π_1 , the probability of alternative outcomes y_i is

$$P(y_i, n_i | \pi_i) = \binom{n_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{n_i - y_i}$$

Similarly, given success proportions π_i specified by our model, the joint log likelihood of

Summary, given success proportions n_i specified by our model, the joint log likelihood of obtaining our observations y_i on all N conditions is

$$\begin{aligned} LL(a_0, a_1, \dots) &= \ln \prod_i = \sum_i \log \left[\binom{n_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{n_i - y_i} \right] = \\ &= \sum_i \log \left[\binom{n_i}{y_i} \right] + \sum_i y_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + \sum_i n_i \log(1 - \pi_i) \\ &= \sum_i \log \left[\binom{n_i}{y_i} \right] + \sum_i y_i \lambda_i - \sum_i n_i \log(1 + e^{\lambda_i}) \end{aligned}$$

In the last line, we replaced the model probabilities π_i with the corresponding logit parameters λ_i .

To perform maximum likelihood fitting of linear coefficients $a_0, a_1, \dots$ we set the partial derivatives to zero

$$0 \stackrel{!}{=} \frac{\partial LL}{\partial a_0}, \quad 0 \stackrel{!}{=} \frac{\partial LL}{\partial a_1}, \quad \dots$$

$$\begin{aligned} 0 \stackrel{!}{=} \frac{\partial LL(a_0, a_1, \dots)}{\partial a_k} &= \sum_i \frac{\partial}{\partial \lambda_i} [y_i \lambda_i - n_i \log(1 + e^{\lambda_i})] \frac{\partial \lambda_i}{\partial a_k} = \\ &= \sum_i \left[y_i - n_i \frac{e^{\lambda_i}}{1 + e^{\lambda_i}} \right] \frac{\partial \lambda_i}{\partial a_k} = \\ &= \sum_i \left[y_i - n_i \frac{e^{\lambda_i}}{1 + e^{\lambda_i}} \right] x_{ik} \quad \forall k \end{aligned}$$

or

$$0 = \sum_i (y_i - n_i \pi_i) x_{ik} \quad \forall k$$

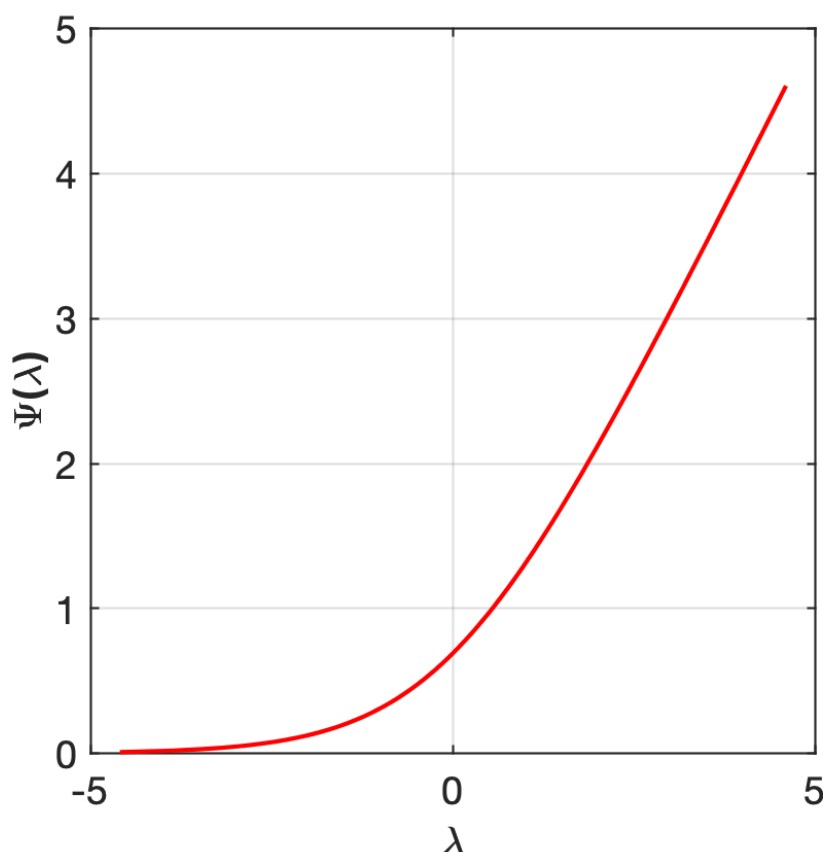
in other words, the predictor-weighted differences between observed fractions n_i and predicted fractions $N_i \pi_i$ must sum to zero. The N predicted fractions $\pi_i = \pi_i(a_k, x_{ik})$ are functions of unknown parameters $a_0, a_1, \dots$ and known predictors x_{ki} .

This means, of course, that the solution for $a_0, a_1, \dots$ must be obtained numerically.

Sufficient statistic (optional)

We can gain some further insight by re-expressing the joint likelihood of observations with a helper function $\Psi(\lambda_i)$

$$\Psi(\lambda_i) = \ln(1 + e^{\lambda_i}) \quad \Leftrightarrow \quad e^{\Psi(\lambda_i)} = 1 + e^{\lambda_i}$$



The binomial likelihood of observing trial outcome y_i , given prediction π_i , is

$$\begin{aligned} \binom{n_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{n_i - y_i} &= \binom{n_i}{y_i} \left(\frac{\pi_i}{1 - \pi_i} \right)^{y_i} (1 - \pi_i)^{n_i} = \\ &= \binom{n_i}{y_i} e^{\lambda_i y_i} [1 + e^{\lambda_i}]^{-n_i} = \\ &= \binom{n_i}{y_i} \exp[\lambda_i y_i - n_i \Psi(\lambda_i)] \end{aligned}$$

The joint likelihood of observations $\mathbf{y}$ as a function of predictors $\mathbf{x}$ and the to-be-optimized parameters $(a_0, a_1, \dots)$ and

$$\begin{aligned} L_{(a_0, a_1, \dots)}(\mathbf{y}) &= \prod_{i=1}^N \binom{n_i}{y_i} \exp[\lambda_i y_i - n_i \Psi(\lambda_i)] = \\ &= \exp \left[\sum_i \lambda_i y_i \right] \exp \left[- \sum_i \Psi(\lambda_i) n_i \right] \prod_{i=1}^N \binom{n_i}{y_i} \end{aligned}$$

With

$$\exp \left[\sum_i \lambda_i y_i \right] = \exp \left[\sum_i (a_0 + a_1 x_{i1} + \dots) y_i \right] = \exp \left[a_0 \sum_i y_i + a_1 \sum_i y_i x_i + \dots \right]$$

we obtain

$$L_{(a_0, a_1, \dots)}(\mathbf{y}) = \exp[a_0 S_0 + a_1 S_1 + \dots] \cdot \exp\left[-\sum_i \Psi(a_0 + a_1 x_i + \dots) n_i\right] \cdot \prod_{i=1}^N \binom{n_i}{y_i}$$

Only the first of these three factors depends on **both** observations $\mathbf{y}$ and predictors $\mathbf{x}$. This implies that $S_0, S_1, S_2, \dots$ are **sufficient statistics** for optimizing $(a_0, a_1, a_2, \dots)$.

The second factor depends only on predictors $\mathbf{x}$, and the third factor depends only on observations $\mathbf{y}$.

The implication is that the numbers $S_0, S_1, S_2, \dots$ alone contain all of the experiment's information. It is the logistic parametrization that makes this happen.

Cell infusion example

In this example, human muscle cell colonies were infused with mouse nuclei in five different ratios, cultured over time periods ranging from one to five days, and observed to see whether they thrived. For example, of the 126 colonies having the third ratio and shortest time period, 48 thrived.

Table 8.2 *Cell infusion data; human cell colonies infused with mouse nuclei in five ratios over 1 to 5 days and observed to see whether they did or did not thrive. Green numbers are estimates $\hat{\pi}_{ij}$ from the logistic regression model. For example, 5 of 31 colonies in the lowest ratio/days category thrived, with observed proportion $5/31 = 0.16$, and logistic regression estimate $\hat{\pi}_{11} = 0.11$.*

		Time				
		1	2	3	4	5
Ratio	1	5/31 .11	3/28 .25	20/45 .42	24/47 .54	29/35 .75
	2	15/77 .24	36/78 .45	43/71 .64	56/71 .74	66/74 .88
	3	48/126 .38	68/116 .62	145/171 .77	98/119 .85	114/129 .93
	4	29/92 .32	35/52 .56	57/85 .73	38/50 .81	72/77 .92
	5	11/53 .18	20/52 .37	20/48 .55	40/55 .67	52/61 .84

Let π_{ij} denote the (unknown) true probability of thriving for ratio i during time period j , and

λ_{ij} the **logits**

$$\lambda_{ij} = \log \frac{\pi_{ij}}{1 - \pi_{ij}}$$

Our model for λ_{ij} is a two-additive regression, with α s changing over times and β s changing over ratios, as follows

$$\lambda_{ij} = \mu + \alpha_i + \beta_j, \quad i, j = 1, 2, \dots, 5$$

with constraints

$$\sum_i \alpha_i = \sum_j \beta_j = 0$$

to avoid redundancies. Accordingly, we have nine free parameters overall (four α 's, four β 's, and μ).

The binomial standard error of the 'thriving' proportions is

$$SE_{\pi} = \sqrt{\frac{\pi_i(1 - \pi_i)}{n_i}}$$

based on the observed proportions π_i (i.e., considering all conditions separately), and

$$SE_{\hat{\pi}} = \sqrt{\frac{\hat{\pi}_i(1 - \hat{\pi}_i)}{n_i}},$$

base on the fitted proportions $\hat{\pi}_i$ (i.e., considering all conditions jointly).

✓ Practicalities

The design matrix $\mathbf{X} = x_{ij,k}$ is of size (25×9)

$$\begin{pmatrix} x_{11,1} & x_{11,2} & \dots & x_{11,5} & x_{11,6} & \dots & x_{11,9} \\ x_{21,1} & x_{21,2} & \dots & x_{21,5} & x_{21,6} & \dots & x_{21,9} \\ x_{31,1} & x_{31,2} & \dots & x_{31,5} & x_{31,6} & \dots & x_{31,9} \\ x_{41,1} & x_{41,2} & \dots & x_{41,5} & x_{41,6} & \dots & x_{41,9} \\ x_{51,1} & x_{51,2} & \dots & x_{51,5} & x_{51,6} & \dots & x_{51,9} \\ x_{12,1} & x_{12,2} & \dots & x_{12,5} & x_{12,6} & \dots & x_{12,9} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{15,1} & x_{15,2} & \dots & x_{15,5} & x_{15,6} & \dots & x_{15,9} \\ x_{25,1} & x_{25,2} & \dots & x_{25,5} & x_{25,6} & \dots & x_{25,9} \\ x_{35,1} & x_{35,2} & \dots & x_{35,5} & x_{35,6} & \dots & x_{35,9} \\ x_{45,1} & x_{45,2} & \dots & x_{45,5} & x_{45,6} & \dots & x_{45,9} \end{pmatrix} = \begin{pmatrix} 1 & 1 & \dots & 0 & 1 \\ 1 & 0 & \dots & 0 & 1 \\ 1 & 0 & \dots & 0 & 1 \\ 1 & 0 & \dots & 0 & 1 \\ 1 & -1 & \dots & -1 & 1 \\ 1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} \sim_{40,1} & \sim_{40,2} & \dots & \sim_{40,5} & \sim_{40,6} & \dots & \sim_{40,9} \\ x_{55,1} & x_{55,2} & \dots & x_{55,5} & x_{55,6} & \dots & x_{55,9} \end{pmatrix} \begin{pmatrix} 1 & -1 & \dots & -1 & -1 \end{pmatrix}$$

Multiplication with the parameter vector β gives

$$\lambda_{ij} = \begin{pmatrix} \mu + \alpha_1 + \beta_1 \\ \mu + \alpha_2 + \beta_1 \\ \mu + \alpha_3 + \beta_1 \\ \mu + \alpha_4 + \beta_1 \\ \mu - \alpha_1 - \alpha_2 - \alpha_3 - \alpha_4 + \beta_1 \\ \mu + \alpha_1 + \beta_2 \\ \vdots \\ \mu + \alpha_1 - \beta_1 - \beta_2 - \beta_3 - \beta_4 \\ \mu + \alpha_2 - \beta_1 - \beta_2 - \beta_3 - \beta_4 \\ \mu + \alpha_3 - \beta_1 - \beta_2 - \beta_3 - \beta_4 \\ \mu + \alpha_4 - \beta_1 - \beta_2 - \beta_3 - \beta_4 \\ \mu - \alpha_1 - \alpha_2 - \alpha_3 - \alpha_4 - \beta_1 - \beta_2 - \beta_3 - \beta_4 \end{pmatrix} = \begin{pmatrix} 1 & 1 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 1 & -1 & \dots & - \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 1 \\ 1 & -1 & \dots & - \end{pmatrix}$$

The joint binomial likelihood of trial outcomes y_i , given predictions π_i , is

$$\begin{aligned} \prod_i \binom{n_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{n_i - y_i} &= \prod_i \binom{n_i}{y_i} \left(\frac{\pi_i}{1 - \pi_i} \right)^{y_i} (1 - \pi_i)^{n_i} = \\ &= \prod_i \binom{n_i}{y_i} [e^{\lambda_i}]^{y_i} [1 + e^{\lambda_i}]^{-n_i} = \end{aligned}$$

and the negative logarithm is

$$-\sum_i \log \binom{n_i}{y_i} - \sum_i y_i \lambda_i + n_i \sum_i \log(1 + e^{\lambda_i})$$

The parameter-dependent terms that need to be considered in the minimization are

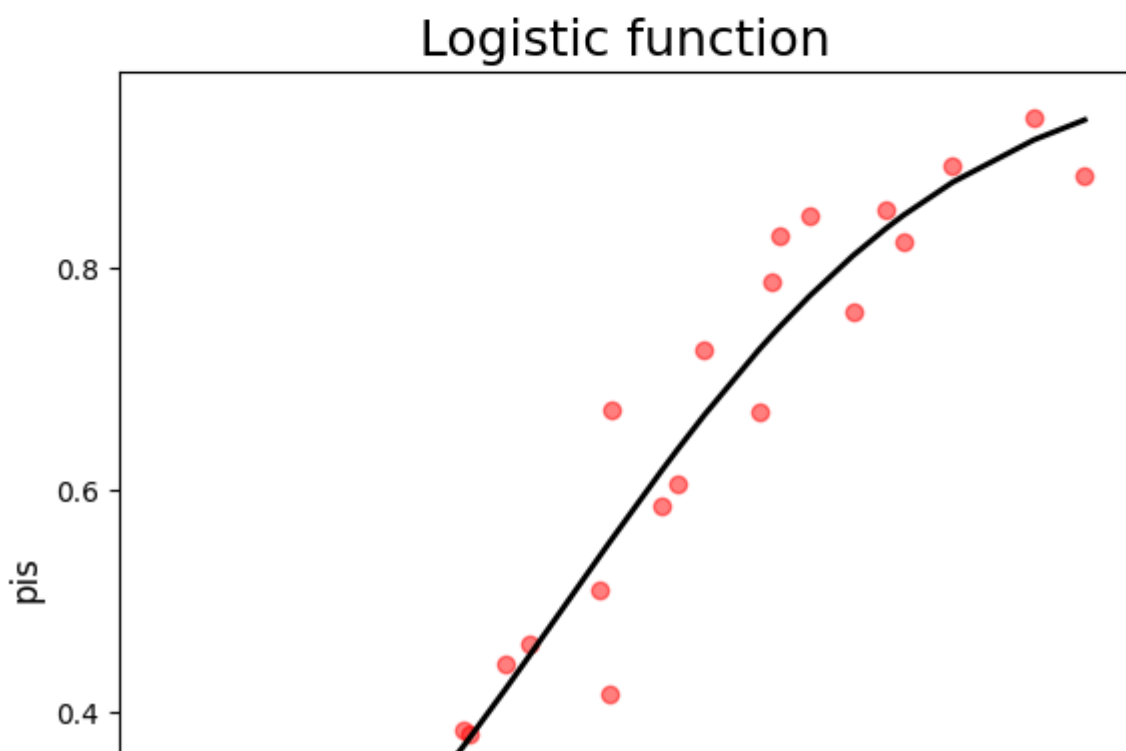
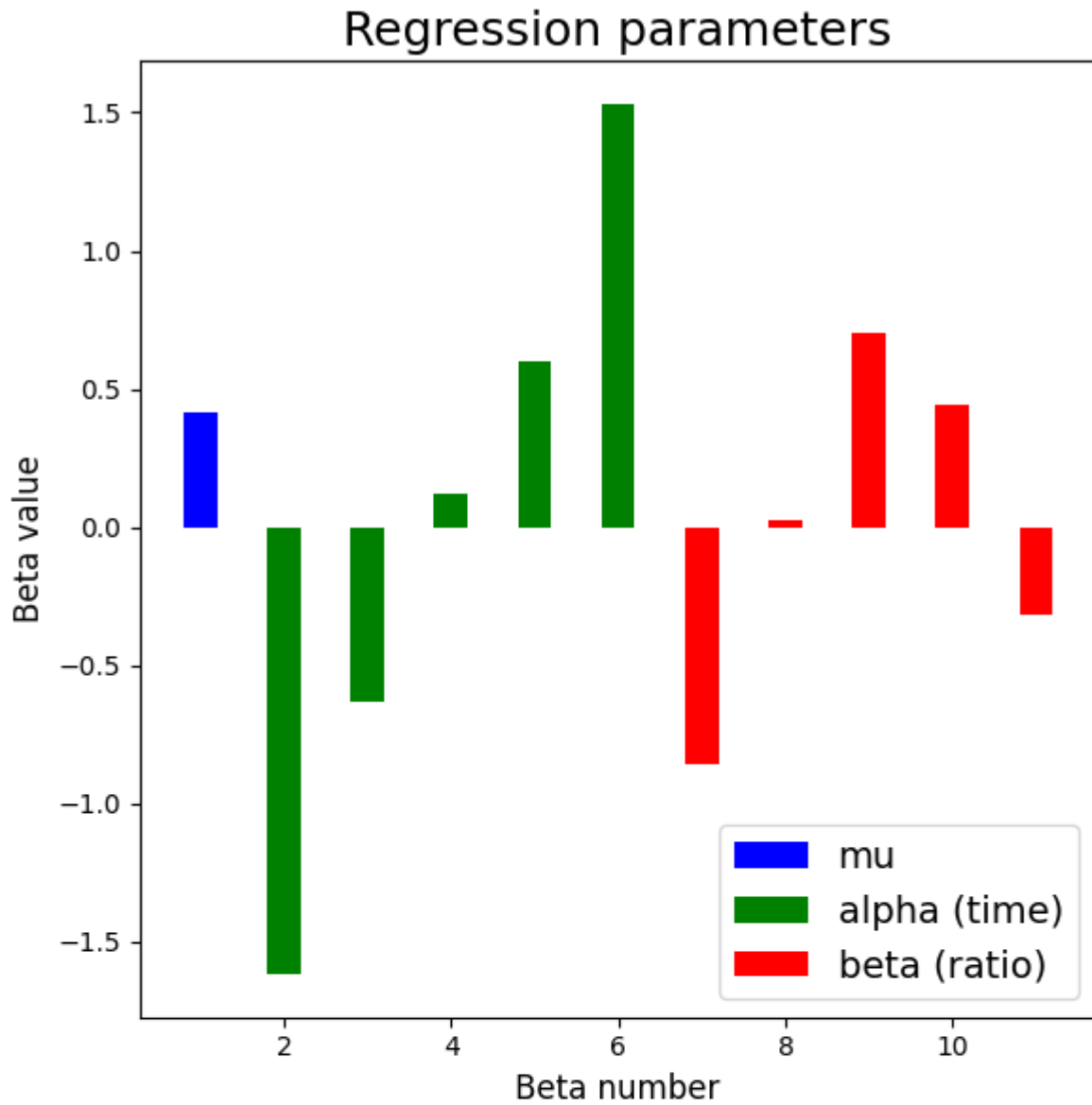
$$LL = -\sum_i [y_i \lambda_i - n_i \log(1 + e^{\lambda_i})]$$

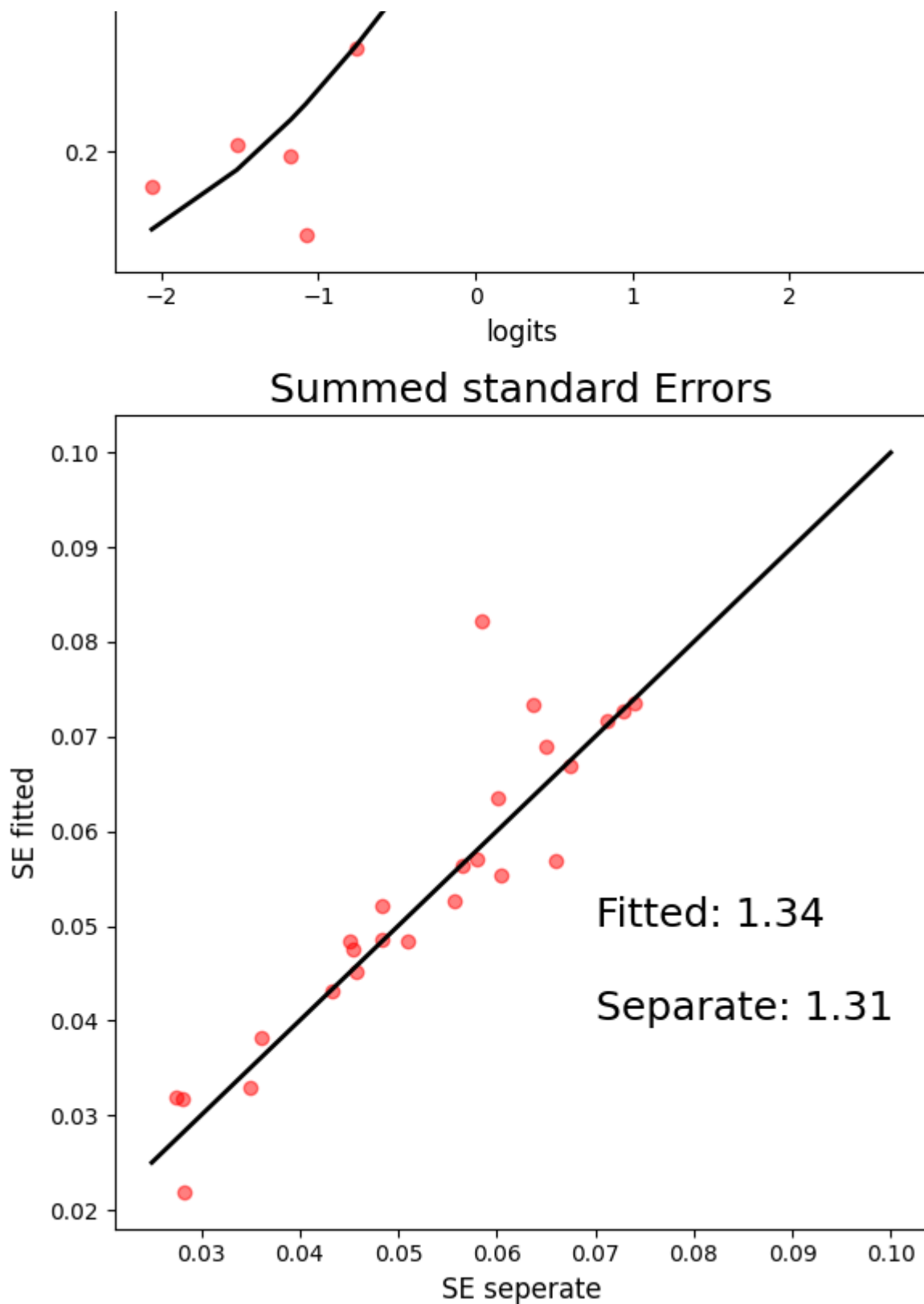
Logistic regression, cell infusion example

Show code

```
Estimated pi's: [[0.11295619 0.25360615 0.41995125 0.53924515 0.74728736]
 [0.23573104 0.45145221 0.63684408 0.73922999 0.87748892]
 [0.37716653 0.61770798 0.77492429 0.84769117 0.93360916]
 [0.31857136 0.55504558 0.72662694 0.81120416 0.9156567 ]]
```

```
[0.17953804 0.36863968 0.55439399 0.6679035 0.83556746]]  
Estimated beta's: [ 0.41281869 -1.61652678 -0.63510338 0.12139407 0.601  
0.0274798 0.70211573 0.44336307]
```





4. Generalized linear models

Logistic regression is a special case of Generalized Linear Models (GLMs). GLMs extend ordinary linear regression, that is least squares curve-fitting, to situations where the response variables are binomial, Poisson, gamma, beta, or in fact any exponential family form.

A one-parameter distribution in the exponential family is

$$\left\{ f_{\lambda}(y) = e^{\lambda y - \gamma(y)} f_0(y), \quad \lambda \in \Lambda \right\}$$

where y is a data point from our experiment, λ is the **natural parameter** of the distribution, $\gamma(\lambda)$ is the **normalizing function**, and $f_0(y)$ is the distribution for $\lambda = 0$. For example, in case of the normal distribution, where σ is assumed to be known

$$f_{\lambda}(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right] = \exp\left[\frac{\mu y}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right] f_{\lambda_0}(y) = \exp[\lambda y - \gamma(y)] f_{\lambda_0}$$

where we have defined

$$\lambda = \frac{\mu}{\sigma^2}, \quad \gamma(\lambda) = \frac{\sigma^2 \lambda^2}{2} = \frac{\mu^2}{2\sigma^2}, \quad f_0(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y^2}{2\sigma^2}\right)$$

We assume that each point from our data set $\mathbf{y} = (y_1, y_2, \dots, y_N)$ independently samples members from this family of distributions, with corresponding natural parameters

$$\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_N)$$

$$y_i \sim f_{\lambda}(\lambda_i), \quad \text{independently } i = 1, 2, \dots, N$$

Table 8.4 Exponential family form for first four cases in Table 5.1; natural parameter λ , sufficient statistic y , deviance (8.31) between family members f_1 and f_2 , $D(f_1, f_2)$, and normalizing function $\gamma(\lambda)$.

	λ	y	$D(f_1, f_2)$	$\gamma(\lambda)$
1. Normal $\mathcal{N}(\mu, \sigma^2)$, σ^2 known	μ/σ^2	x	$\left(\frac{\mu_1 - \mu_2}{\sigma}\right)^2$	$\sigma^2 \lambda^2 / 2$
2. Poisson $\text{Poi}(\mu)$	$\log \mu$	x	$2\mu_1 \left[\left(\frac{\mu_2}{\mu_1} - 1\right) - \log \frac{\mu_2}{\mu_1} \right]$	e^{λ}
3. binomial $\text{Bi}(n, \pi)$	$\log \frac{\pi}{1-\pi}$	x	$2n \left[\pi_1 \log \frac{\pi_1}{\pi_2} + (1 - \pi_1) \log \frac{1-\pi_1}{1-\pi_2} \right]$	$n \log(1 + e^{\lambda})$
4. Gamma $\text{Gam}(v, \sigma)$, v known	$-1/\sigma$	x	$2v \left[\left(\frac{\sigma_1}{\sigma_2} - 1\right) - \log \frac{\sigma_1}{\sigma_2} \right]$	$-v \log(-\lambda)$

Maximum likelihood fitting

The N natural parameters λ_i are fitted with a linear regression

$$\boldsymbol{\lambda} = \mathbf{X} \cdot \boldsymbol{\alpha}$$

where $\mathbf{X}$ is an $N \times P$ design matrix and $\boldsymbol{\alpha}$ is a $P \times 1$ parameter vector, with $P \ll N$.

The likelihood distribution of the observations, given our model and its parameters, is then

$$L(\boldsymbol{\alpha}) = \prod_i f_{\lambda_i}(y_i) = \exp \left[\sum_i [\lambda_i y_i - \gamma(\lambda_i)] \right] \prod_i f_0(y_i)$$

where the dot-product

$$\sum_i y_i \lambda_i = \mathbf{y} \cdot \underbrace{\mathbf{X} \cdot \boldsymbol{\alpha}}_{\mathbf{z}}$$

includes the sufficient statistic $\mathbf{z} = \mathbf{y} \cdot \mathbf{X}$ and the sum of gamma-terms

$$\sum_i \gamma(\lambda_i) = \sum_i \gamma(\mathbf{x}_i \cdot \boldsymbol{\alpha})$$

involves the dot-products between rows of $\mathbf{X}$ and parameter vector $\boldsymbol{\alpha}$.

The univariate likelihood distributions are characterized by their respective means μ_λ and variances σ_λ^2

$$y \sim (\mu_\lambda, \sigma_\lambda^2)$$

and the multivariate distribution has mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\Sigma}$, which is diagonal due to independence

$$\mathbf{y} \sim (\boldsymbol{\mu}(\boldsymbol{\alpha}), \boldsymbol{\Sigma}(\boldsymbol{\alpha}))$$

The MLE $\hat{\boldsymbol{\alpha}}$ satisfies

$$0 = \mathbf{X}^T [\mathbf{y} - \boldsymbol{\mu}(\hat{\boldsymbol{\alpha}})]$$

and is distributed as with mean and covariance

$$\hat{\boldsymbol{\alpha}} \sim \left[\boldsymbol{\alpha}, (\mathbf{X}^T \cdot \boldsymbol{\Sigma}(\boldsymbol{\alpha}) \cdot \mathbf{X})^{-1} \right]$$

and its numerical solution is facilitated by the convexity of $\mathbf{L}(\boldsymbol{\alpha})$,

The MLE $\hat{\boldsymbol{\alpha}}$ minimizes the total divergence

$$\sum_i D[y_i, \mu_i(\alpha)], \quad D[f_1, f_2] \equiv \int_y f_1(y) \log \frac{f_1(y)}{f_2(y)} dy$$

Conclusion

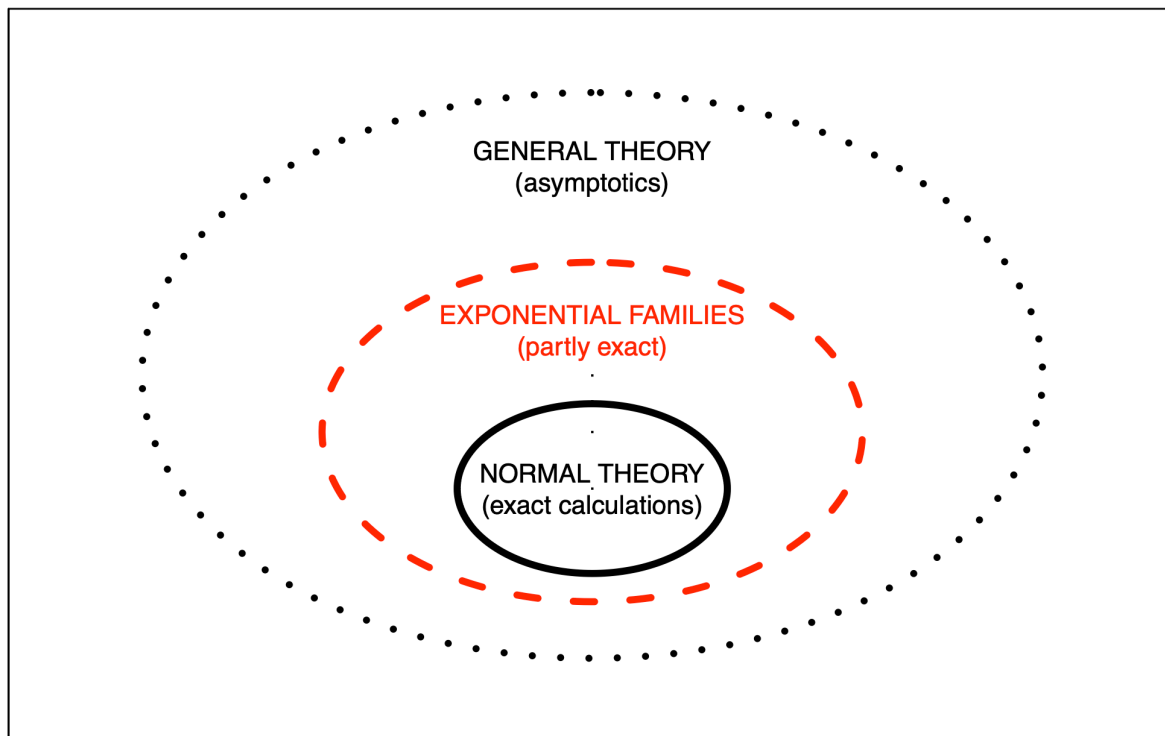


Figure 8.3 Three levels of statistical modeling.

The inner circle of represents parametric statistics of normally distributed variables. Exact inferences—t-tests, F distributions, and multivariate analysis are feasible there.

The middle circle comprises some exact results relating to a few special families: the binomial, Poisson, gamma, beta, and so on. Exponential family theory unifies this middle circle into a coherent whole. Even the approximations are more robust and convincing than in the outer circle ('general theory').

5. Gamma regression

Gamma distributions arise from accumulations of multiple events, each of which occurs with exponentially distributed waiting times.

Examples include:

- Multistable perception: visual, auditory, and somatosensory perceptions and typically unstable, in the sense that they are eventually replaced with alternative perceptions, even when stimulation remains constant. The persistence duration of such perceptions is Gamma distributed.

is Gamma distributed.

- **Gene Expression Levels:** measured as mRNA or protein abundance, due to the stochastic nature of transcription and translation, where bursts of molecular synthesis occur at random intervals.
- **Time to Cell Division:** The time it takes for a cell to complete its cycle and divide (intermitotic time) can be gamma-distributed. This reflects the accumulation of multiple stochastic biochemical steps (e.g., DNA replication, checkpoint signaling) that must occur before division.
- **Enzyme Kinetics:** The time taken for an enzyme to catalyze a reaction or the number of substrate molecules processed in a given time can follow a gamma distribution. This is because enzymatic reactions often involve multiple intermediate steps, each with exponential-like waiting times.
- **DNA Mutation Accumulation:** The number of mutations accumulated in a genome over time, especially in rapidly dividing cells or under mutagenic stress, can follow a gamma distribution due to the independent occurrence of mutation events modulated by repair mechanisms.

✓ Binocular rivalry rexample

We investigated binocular rivalry in human observers at five different image contrasts. This produced five sets of 'perceptual dominance' periods (approximately 800 periods per set). We believe that the different sets sample Gamma distributions with different rate parameters, but all with the same shape parameter $\nu \approx 3.5$.

Let $\mathbf{n}$ be a $(N_{sample} \times 1)$ vector of ones and $\mathbf{c}$ be a $(N_{sample} \times 1)$ vector of contrast values $c = 1, 2, \dots, 5$. Then we can define a design matrix

$$\mathbf{X} = [\mathbf{n} \quad \mathbf{c} \quad \mathbf{n}]$$

for three unknown parameters $\boldsymbol{\alpha} = [\alpha_0, \alpha_1, \alpha_2]$.

The natural parameters $\lambda_{1,2} = (\frac{1}{\theta}, \nu)$ of the Gamma distribution are the inverse scale (or rate) parameter $\frac{1}{\theta}$ and shape parameter ν , which we fit as

$$\lambda_1 = X_0\alpha_0 + X_1\alpha_1, \quad \lambda_2 = X_2\alpha_2$$

The Gamma distribution is defined as

$$f_{\lambda}(y) = \frac{1}{\Gamma(\nu)} \frac{y^{\nu-1}}{\theta^{\nu}} \exp\left(-\frac{y}{\theta}\right) = \frac{1}{\Gamma(\nu)} \left[\frac{y^{\nu-1}}{\theta^{\nu}} \exp\left(-\frac{y}{\theta}\right) \right]$$

$$\begin{aligned}
 &= \exp \left[-\frac{1}{\theta} + (\nu - 1) \log y + \nu \log \left(\frac{1}{\theta} \right) - \log \Gamma(\nu) \right] = \\
 &= \exp \left[-\lambda_1 y + (\lambda_2 - 1) \log y + \Psi \right] \\
 \lambda_1 &= \frac{1}{\theta}, \quad \lambda_2 = \nu, \quad \Psi = \lambda_2 \log(\lambda_1) - \log \Gamma(\lambda_2)
 \end{aligned}$$

so that the parameter-dependent part of the log likelihood of observations, given parameters $(\alpha_0, \alpha_1, \alpha_2)$, is

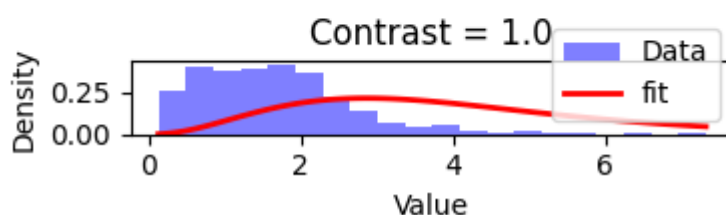
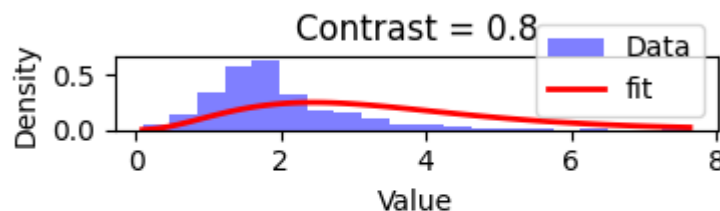
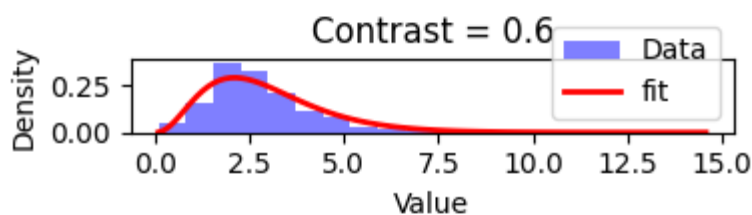
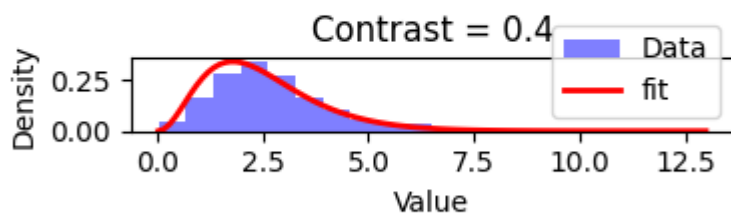
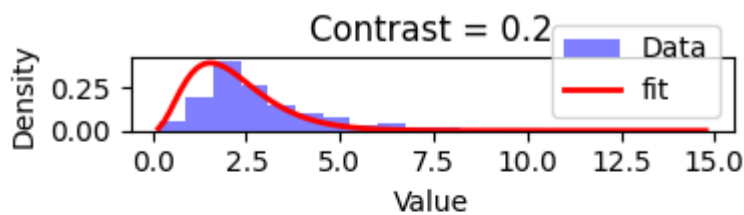
$$LL(\boldsymbol{\alpha}) = \sum_i [-y_i \lambda_{1i} + (\lambda_{2i} - 1) \log y_i + \lambda_{2i} \log(\lambda_{1i}) - \log \Gamma(\lambda_{2i})]$$

The **negative** log-likelihood is what we want to **minimize**.

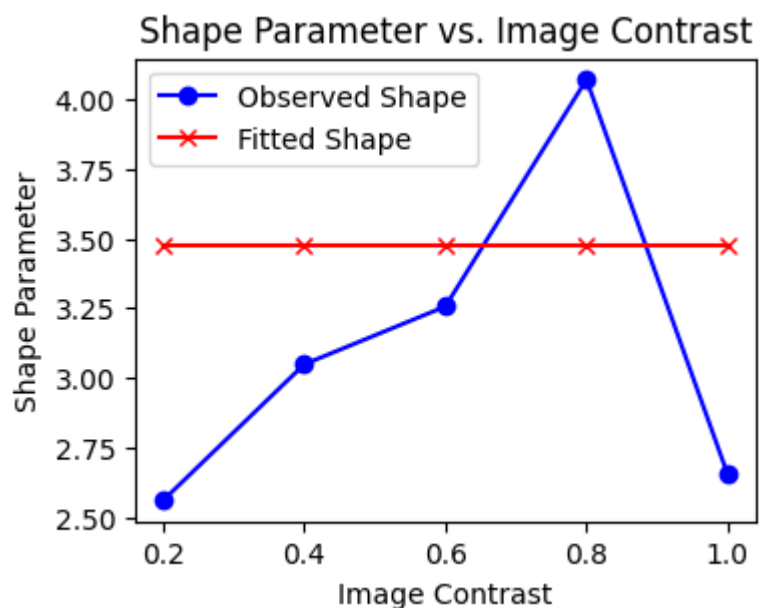
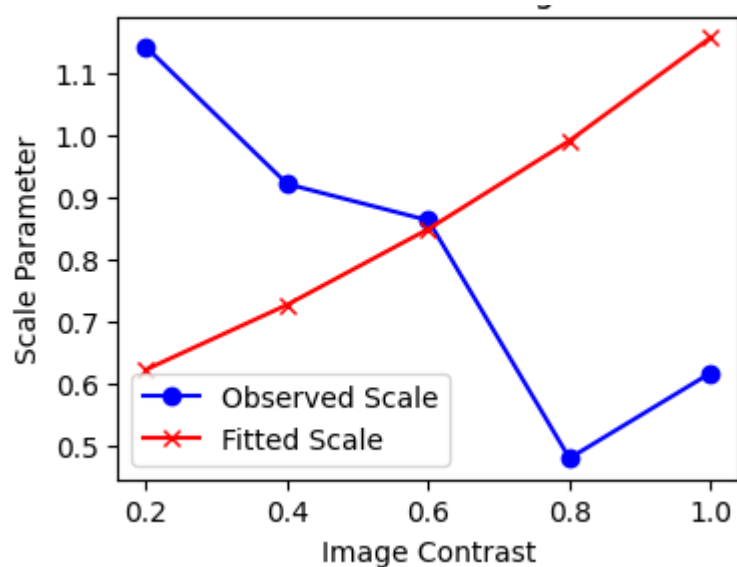
Gamma regression with binocular rivalry data, BUGGY ATTEMPT by hand

Show code

Estimated parameters: [0.61723285 0.77314406 3.5]



Scale Parameter vs. Image Contrast



Gamma regression with binocular rivalry data

Show code

(4397, 2)

GLM Fit Summary:

Generalized Linear Model Regression Results

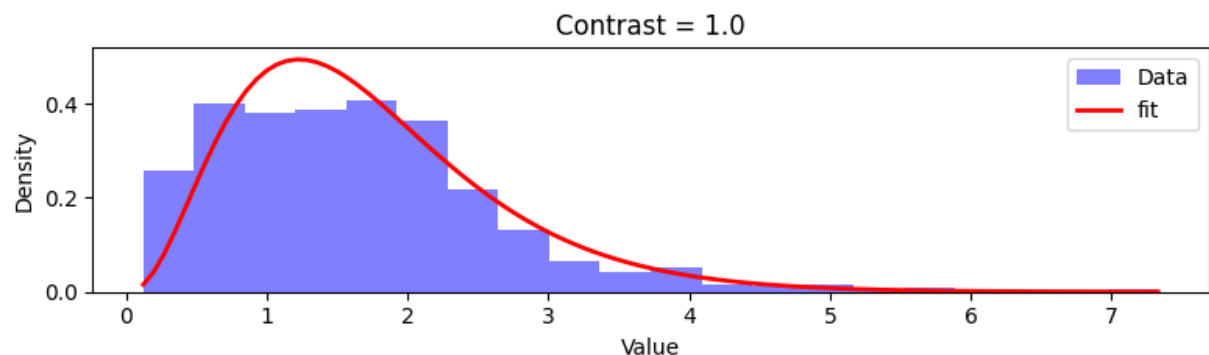
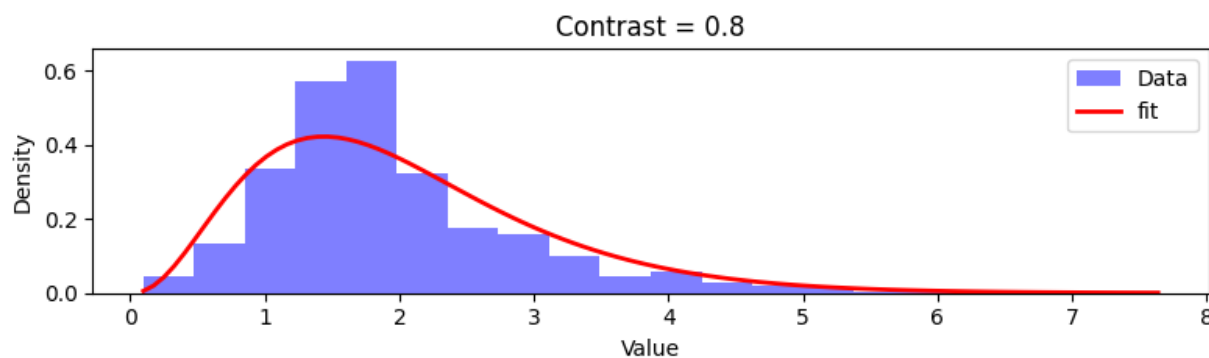
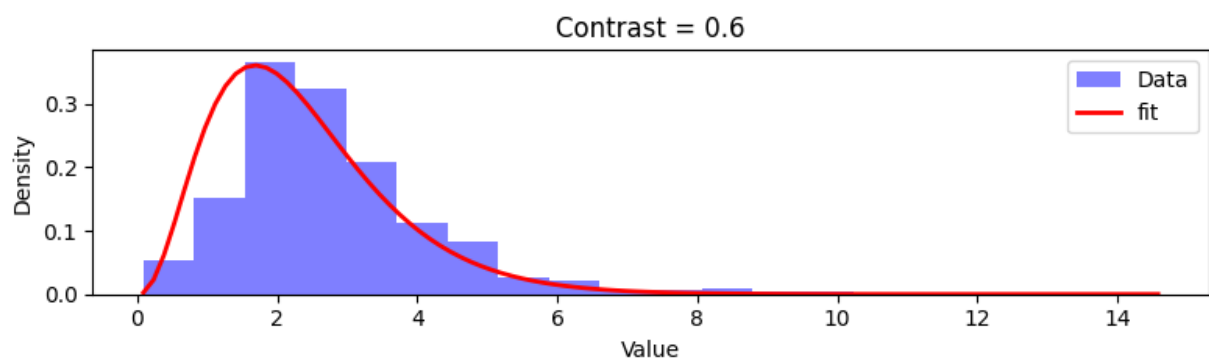
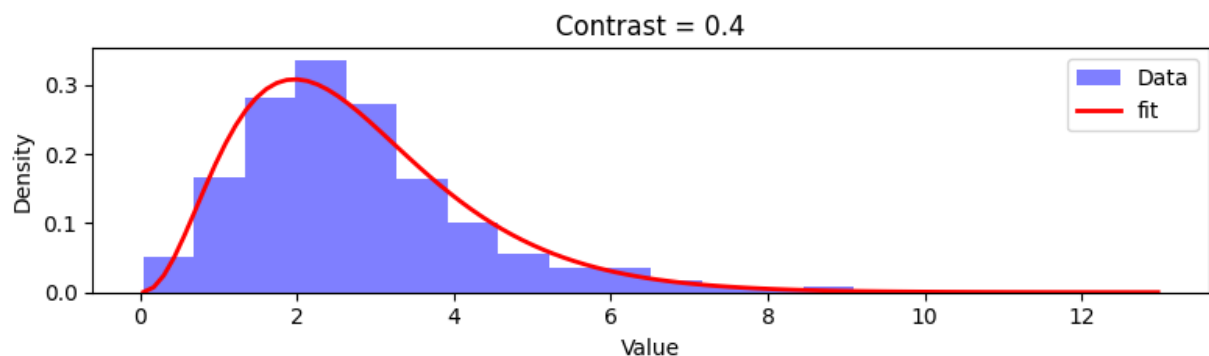
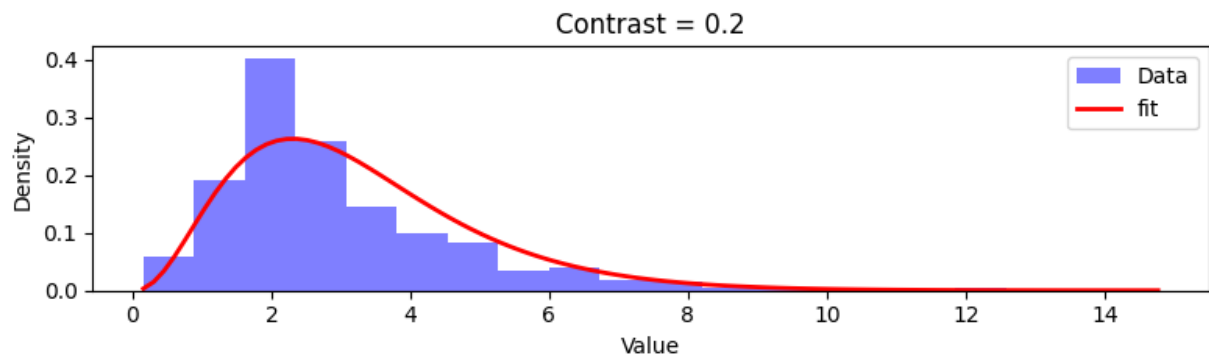
Dep. Variable:	y	No. Observations:	
Model:	GLM	Df Residuals:	
Model Family:	Gamma	Df Model:	
Link Function:	Log	Scale:	0.
Method:	IRLS	Log-Likelihood:	-6
Date:	Tue, 20 May 2025	Deviance:	1
Time:	10:39:43	Pearson chi2:	1.4
No. Iterations:	11	Pseudo R-squ. (CS):	0
Covariance Type:	nonrobust		

	coef	std err	z	P> z	[0.025	0
const	1.3323	0.022	61.429	0.000	1.290	

x1 -0.7872 0.031 -25.267 0.000 -0.848 -

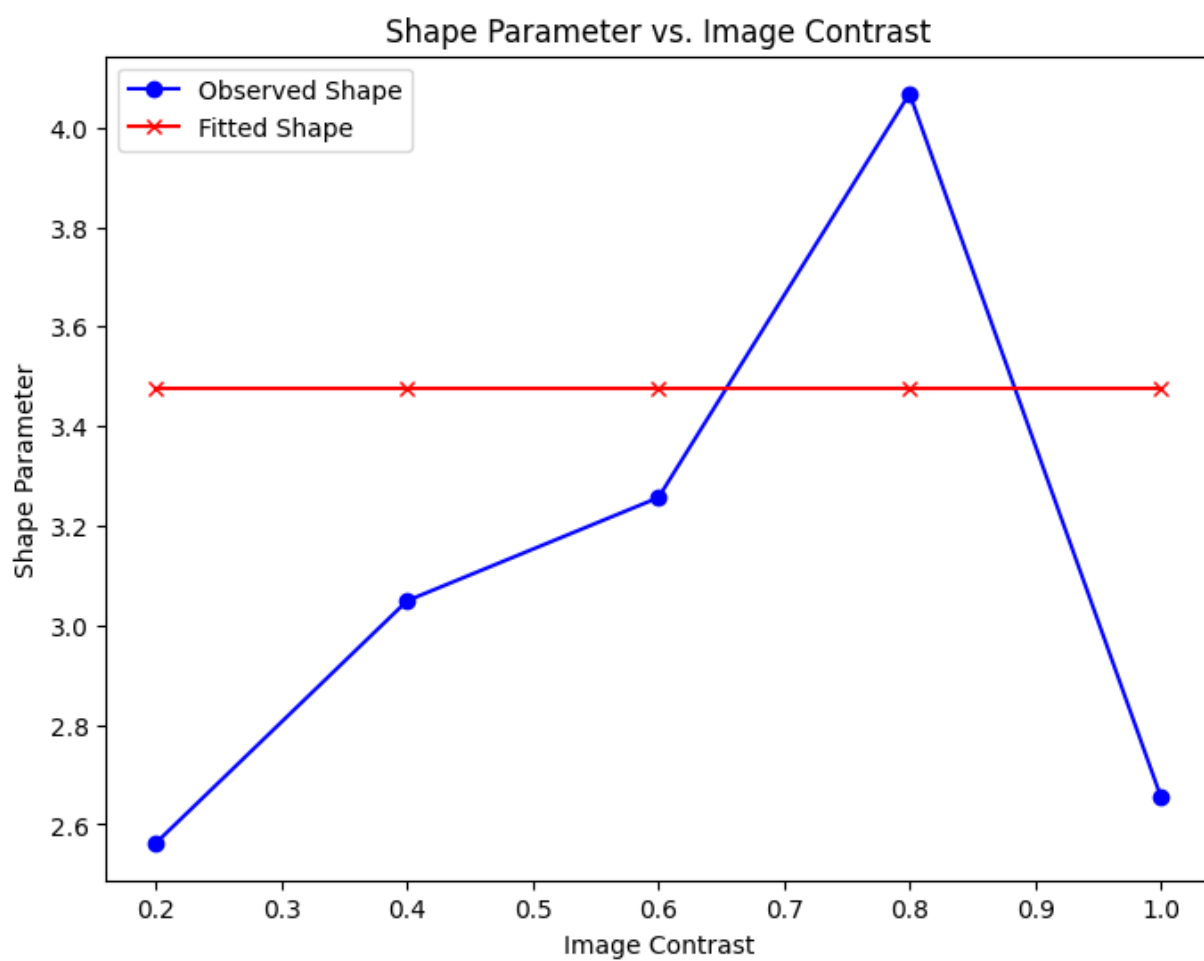
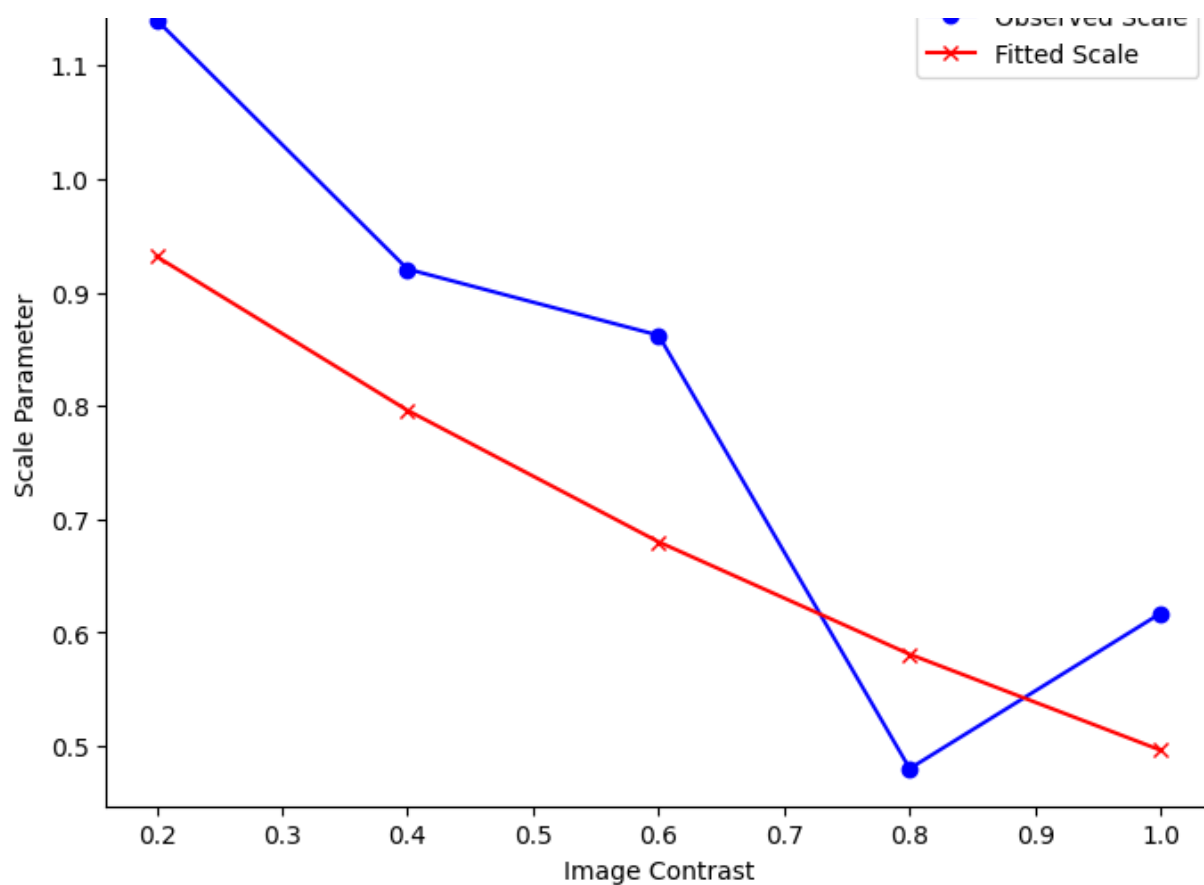
=====

Estimated parameters: 1.3322902366557052 -0.7871745850767088



Scale Parameter vs. Image Contrast





6. Poisson regression

A final useful member of the exponential family is the Poisson distribution, also known as the **law of rare events**. It applies to events that occur independently with a certain average rate, and specifies the probability that we observe 0, 1, 2, . . . events during a given period. The distribution ranges over all positive integers, from zero to infinity. In other words, the Poisson distribution is the limiting case of a Binomial distribution when the number of Bernoulli trials goes to infinity.

A common application of Poisson regression are situations with truncated normal distributions.

Galaxy count example

In this example, we are given the redshift (distance) r and the magnitude (dimness) m of galaxies that are observable in small portion of the sky. A total of 486 galaxies are sorted into 15 redshift bins and 18 magnitude bins, so 270 bins total.

We assume that observed counts are Poisson distributed about an expectation. We hypothesize that the logarithm of this expectation is a polynomial function of r and m , so that the expectation itself is a multivariate normal function of r and m .

Specifically, we assume that expectation $\lambda(r, m)$ is

$$\lambda(r, m) = \alpha_1 r + \alpha_2 m + \alpha_3 r^2 + \alpha_4 r m + \alpha_5 m^2$$

Our goal is to fit the parameter vector $(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5)$.

Table 8.5 *Counts for a truncated sample of 486 galaxies, binned by redshift and magnitude.*

		redshift (farther) →														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
↑ magnitude (dimmer)	18	1	6	6	3	1	4	6	8	8	20	10	7	16	9	4
	17	3	2	3	4	0	5	7	6	6	7	5	7	6	8	5
	16	3	2	3	3	3	2	9	9	6	3	5	4	5	2	1
	15	1	1	4	3	4	3	2	3	8	9	4	3	4	1	1
	14	1	3	2	3	3	4	5	7	6	7	3	4	0	0	1
	13	3	2	4	5	3	6	4	3	2	2	5	1	0	0	0
	12	2	0	2	4	5	4	2	3	3	0	1	2	0	0	1
	11	4	1	1	4	7	3	3	1	2	0	1	1	0	0	0
	10	1	0	0	2	2	2	1	2	0	0	0	1	2	0	0
	9	1	1	0	2	2	2	0	0	0	0	1	0	0	0	0
	8	1	0	0	0	1	1	0	0	0	0	1	1	0	0	0
	7	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0
	6	0	0	3	1	1	0	0	0	0	0	0	0	0	0	0
	5	0	3	1	1	0	0	0	0	0	0	0	0	0	0	0
	4	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
	3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

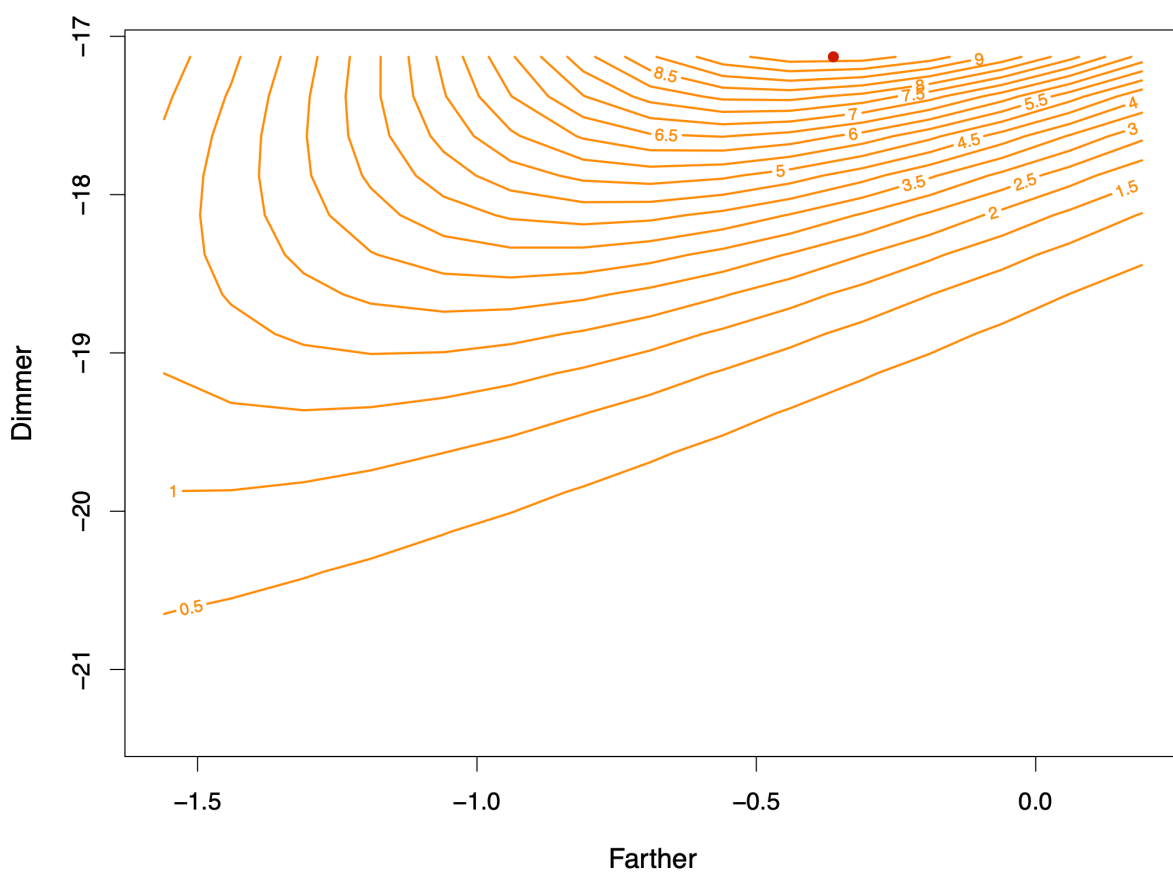


Figure 8.5 Contour curves for Poisson GLM density estimate for the galaxy data. The red dot shows the point of maximum density.

✓ Practicalities

Let $\mathbf{r}$ be a $(15 \times 18 \rightarrow 270 \times 1)$ vector of 18 sets of bin numbers $r = 1, 2, \dots, 15$ and $\mathbf{m}$ the $(15 \times 18 \rightarrow 270 \times 1)$ vector of 15 sets of bin numbers $m = 1, 2, \dots, 18$. Then we can define a design matrix

$$\mathbf{X} = [\mathbf{r} \quad \mathbf{m} \quad \mathbf{r}^2 \quad \mathbf{rm} \quad \mathbf{m}^2]$$

for five unknown parameters $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5]$.

The natural parameter of the Poisson distribution is $\log \mu$, which we fit as

$$\boldsymbol{\lambda}(\boldsymbol{\alpha}) = \log \boldsymbol{\mu}(\boldsymbol{\alpha}) = \mathbf{X} \cdot \boldsymbol{\alpha}$$

The Poisson distribution with natural parameter $\lambda = \log \mu$ is defined as

$$f_{\mu}(y) = \frac{\mu^y e^{-\mu}}{y!} = \frac{e^{y \log \mu - \mu}}{y!} \stackrel{\mu=e^{\lambda}}{=} \frac{e^{y\lambda - e^{\lambda}}}{y!}$$

and the (parameter-dependent part of the) log likelihood of observations, given parameters $\boldsymbol{\alpha}$, is

$$LL(\boldsymbol{\alpha}) = \sum_i [y_i \lambda_i - e^{\lambda_i}]$$

The **negative** log-likelihood is what we want to **minimize**.

Poisson regression, galaxy count example

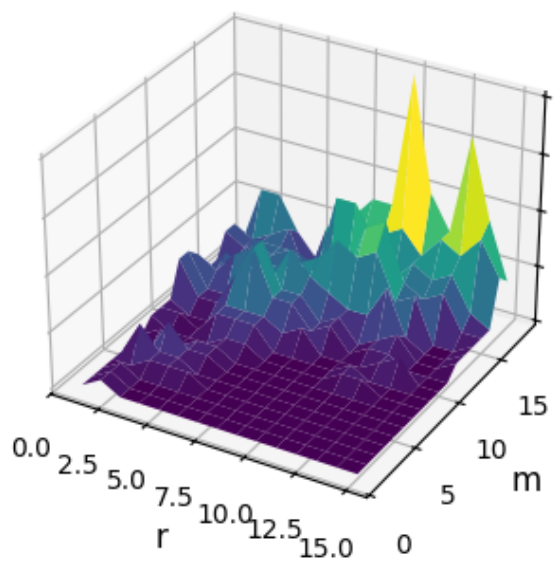
Show code

Generalized Linear Model Regression Results						
=====						
Dep. Variable:	y	No. Observations:				
Model:	GLM	Df Residuals:				
Model Family:	Poisson	Df Model:				
Link Function:	Log	Scale:			1	
Method:	IRLS	Log-Likelihood:			-3	
Date:	Wed, 16 Apr 2025	Deviance:			2	
Time:	15:12:59	Pearson chi2:				
No. Iterations:	7	Pseudo R-squ. (CS):			0	
Covariance Type:	nonrobust					
=====						
	coef	std err	z	P> z	[0.025	0

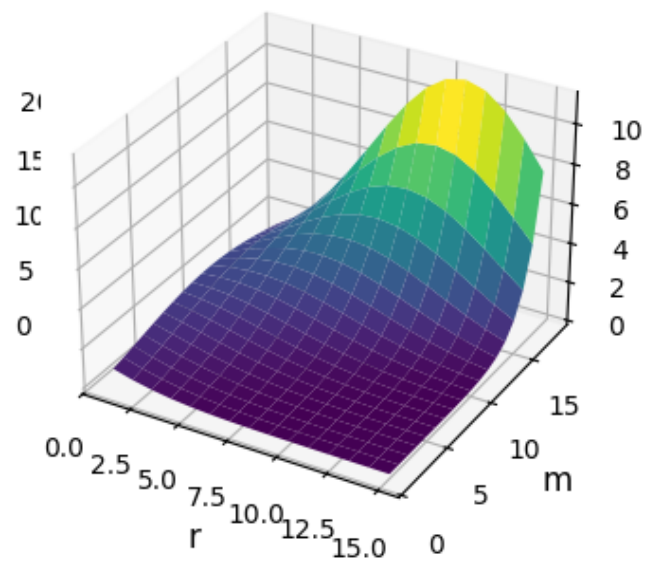
x1	-0.3478	0.064	-5.415	0.000	-0.474	-
x2	0.1419	0.040	3.581	0.000	0.064	
x3	-0.0223	0.003	-6.609	0.000	-0.029	-
x4	0.0464	0.005	9.957	0.000	0.037	
_____	_____	_____	_____	_____	_____	_____

x5	-0.0086	0.0003	-3.340	0.001	-0.014	-
Fitted params (sm):	[-0.34783406	0.14192231	-0.0223069	0.04640992	-0.0	
Fitted params (LL):	[-0.34783406	0.14192231	-0.0223069	0.04640992	-0.0	

Galaxy count



Expectation



Summary:

1. Linear regression
2. Ridge regression
3. Logistic regression
4. Generalized linear models
5. Gamma regression
6. Poisson regression

Following lectures

- Lecture 6: Bayes and bootstrap (Jochen Braun)
- Lecture 7: Bootstrapping examples (Stepan Aleshin)
- Lecture 8: GLM examples (Stepan Aleshin)
- Lecture 9: Linear discriminant analysis (Jochen Braun)
- Lecture 10: Multivariate LDA (Jochen Braun)

